US012403876B2

(12) **United States Patent**
Yanagida et al.

(10) **Patent No.: US 12,403,876 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **VEHICLE BRAKE SYSTEM**

(71) Applicant: **DENSO CORPORATION**, Kariya (JP)

(72) Inventors: **Etsugou Yanagida**, Kariya (JP); **Daisuke Hokuto**, Kariya (JP); **Kunio Nanba**, Kariya (JP)

(73) Assignee: **DENSO CORPORATION**, Kariya (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 536 days.

(21) Appl. No.: **17/835,096**

(22) Filed: **Jun. 8, 2022**

(65) **Prior Publication Data**

US 2022/0297649 A1 Sep. 22, 2022

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2020/046371, filed on Dec. 11, 2020.

(30) **Foreign Application Priority Data**

Dec. 13, 2019 (JP) 2019-225659

(51) **Int. Cl.**
B60T 8/32 (2006.01)
B60T 7/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B60T 8/326*** (2013.01); ***B60T 7/042*** (2013.01); ***B60T 8/172*** (2013.01); ***B60T 11/103*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC B60T 8/326; B60T 17/221; B60T 7/042; B60T 8/172; B60T 8/171; B60T 13/686;
(Continued)

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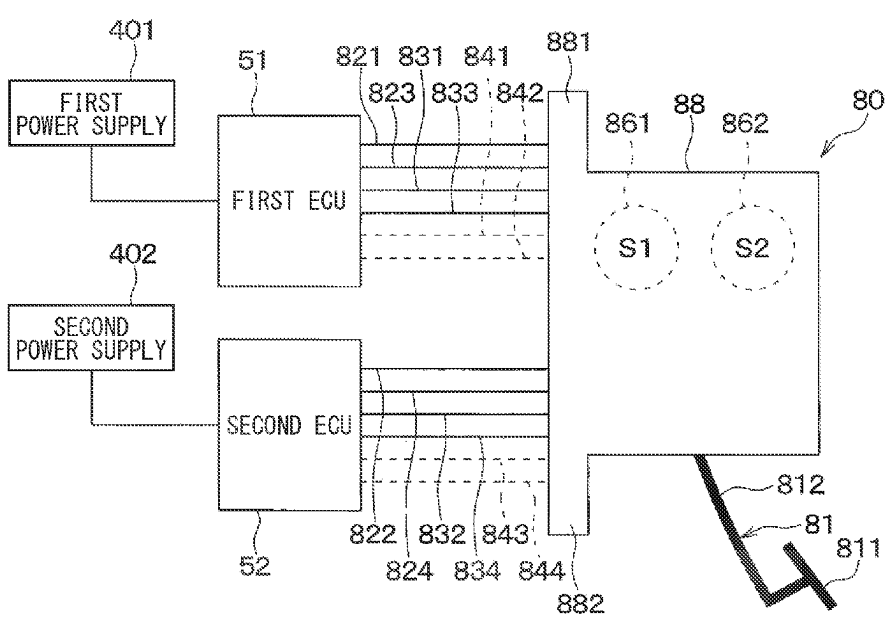
Primary Examiner — Pamela Rodriguez
(74) *Attorney, Agent, or Firm* — Nixon & Vanderhye P.C.

(57) **ABSTRACT**

A vehicle brake system includes a vehicle brake device and a hydraulic pressure generation unit. The vehicle brake device includes: a brake pedal that has a pedal portion, and a lever portion rotating about a rotation axis in response to the pedal portion being operated; a stroke sensor that outputs a signal based on a stroke amount of the brake pedal; a housing that rotatably supports the lever portion; and a reaction force generation portion that generates a reaction force applied on the lever portion based on the stroke amount. The hydraulic pressure generation unit generates a hydraulic pressure for braking a vehicle.

2 Claims, 22 Drawing Sheets



(51) **Int. Cl.**
B60T 8/172 (2006.01)
B60T 11/10 (2006.01)
B60T 13/68 (2006.01)
B60T 17/22 (2006.01)

(52) **U.S. Cl.**
CPC ***B60T 13/686*** (2013.01); ***B60T 17/221*** (2013.01); *B60T 2270/402* (2013.01); *B60T 2270/413* (2013.01)

(58) **Field of Classification Search**
CPC B60T 11/103; B60T 2270/402; B60T 2270/413; B60T 13/662; B60T 2220/04
See application file for complete search history.

(56) **References Cited**

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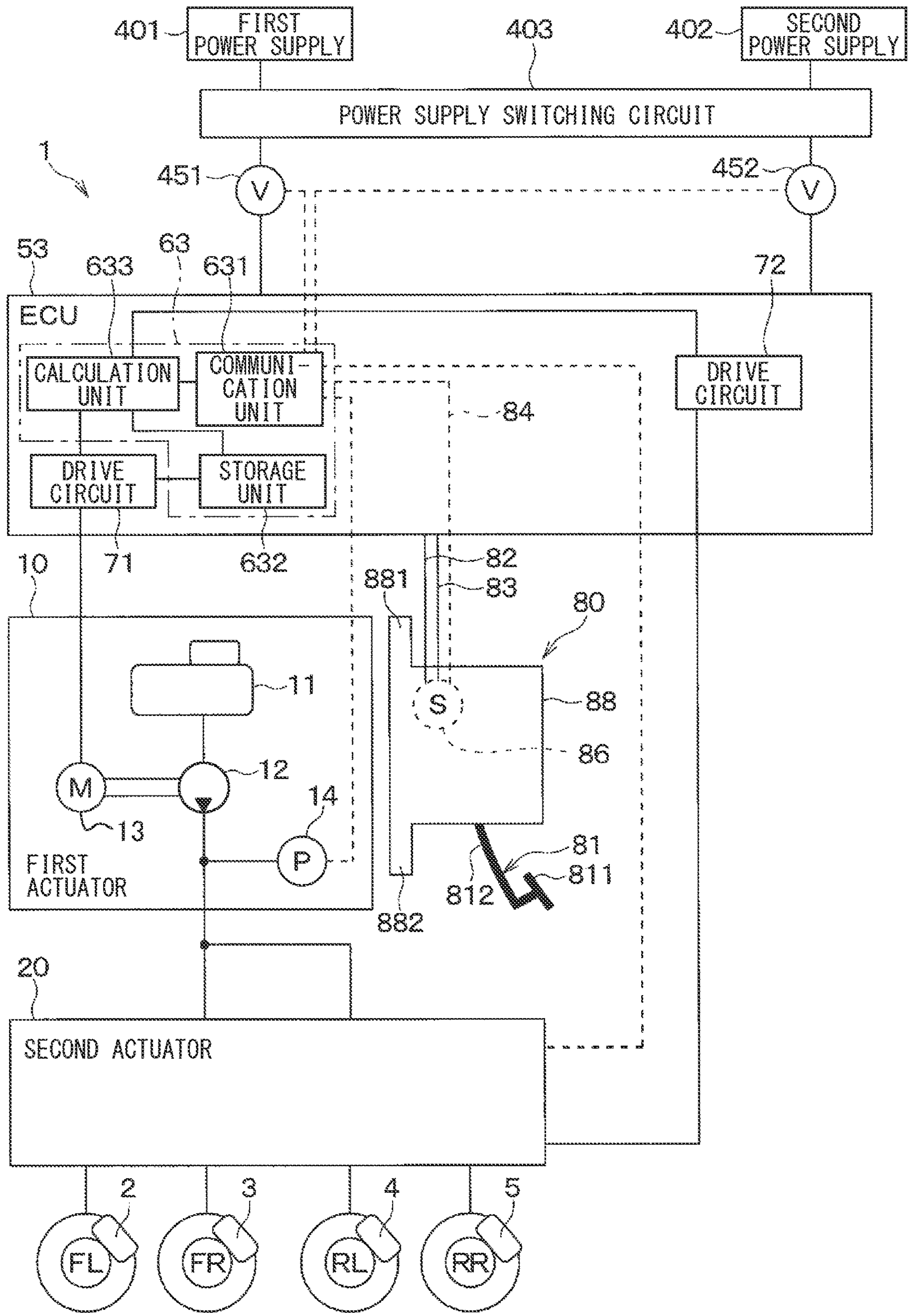
* cited by examiner

FIG. 1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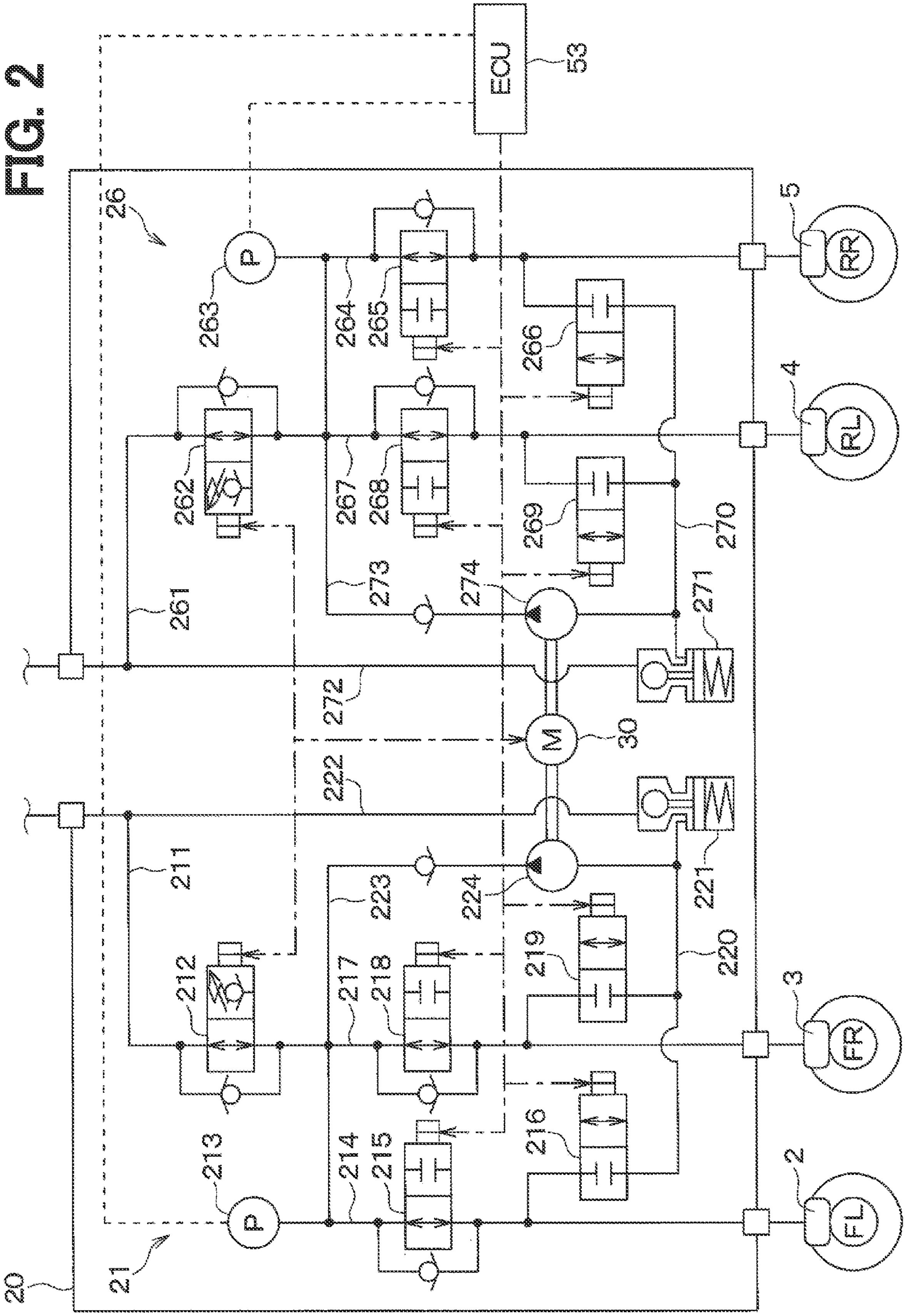
FIRST POWER SUPPLY
403
402
SECOND POWER SUPPLY
POWER SUPPLY SWITCHING CIRCUIT
1
451
V
452
V
53
633
63
631
72
ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
DRIVE CIRCUIT
84
DRIVE CIRCUIT
STORAGE UNIT
10
71
632
82
881
83
80
11
S
88
M
12
86
14
P
13
81
FIRST ACTUATOR
812
811
882
20
SECOND ACTUATOR
2
3
4
5
FL
FR
RL
RR

FIG. 2


ECU
53
20
21
26
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213
263
M
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FL
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RL
RR
2
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4
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211
212
214
215
216
217
218
219
220
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269
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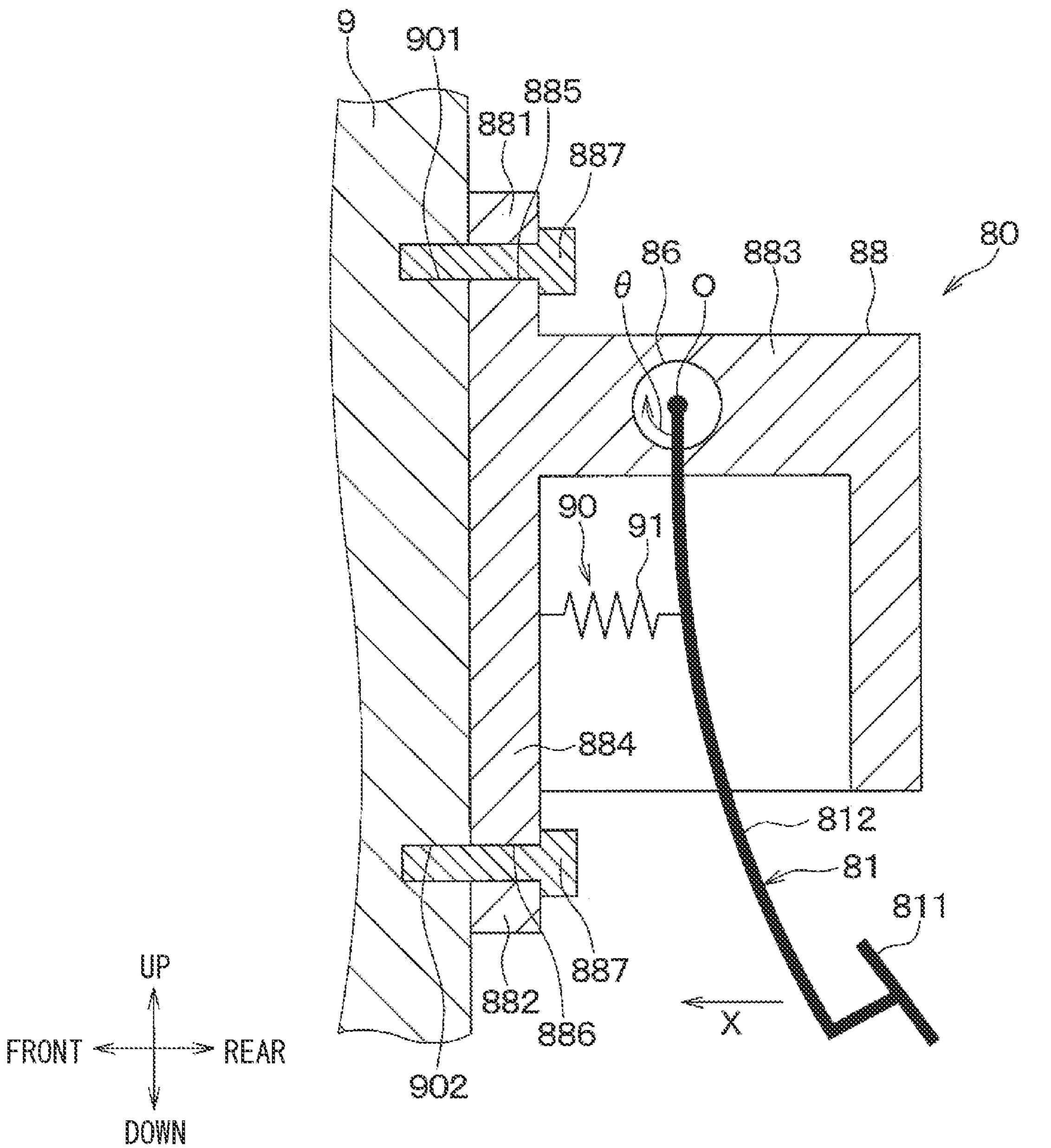
271
272
273
274

FIG. 3



9
901
885
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887
80
86
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884
812
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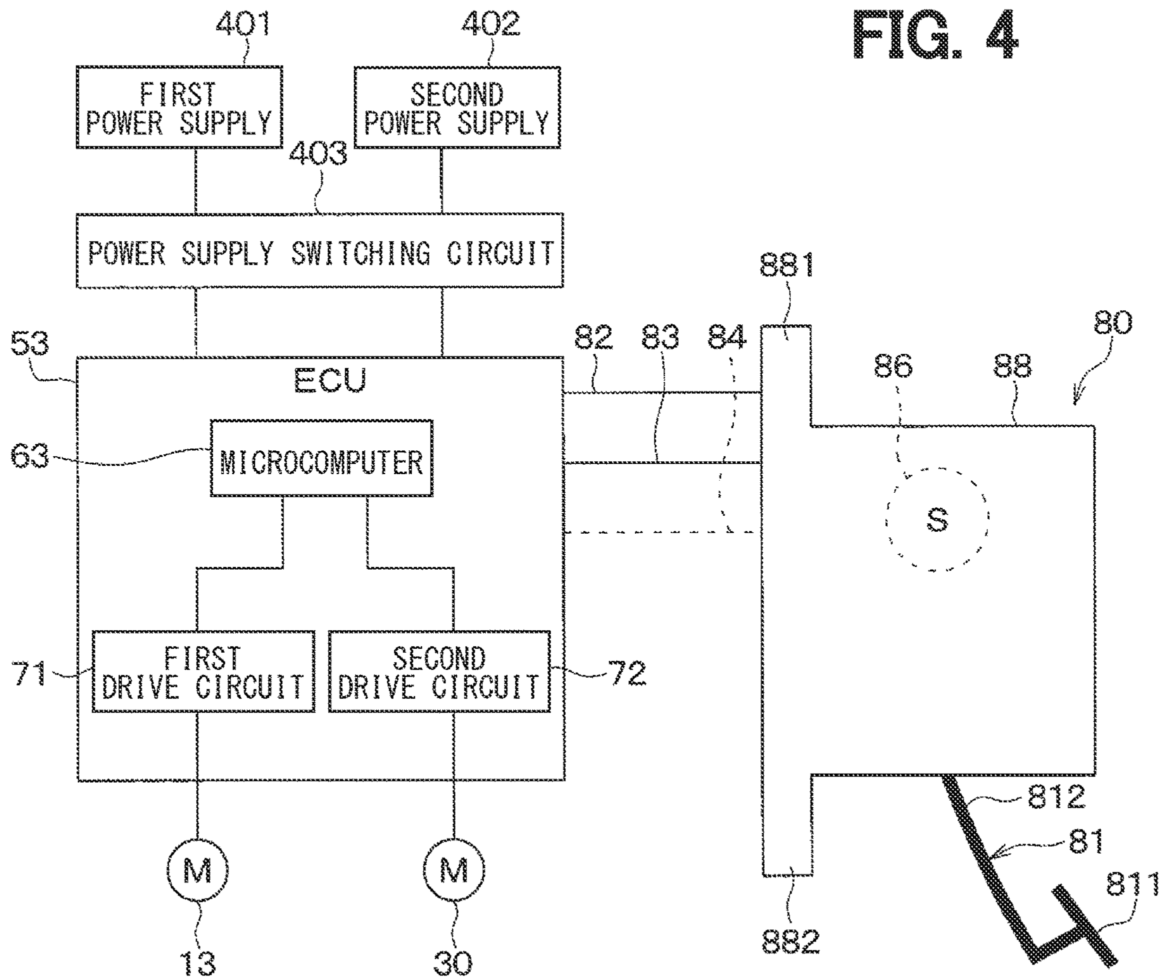
902
X
UP
FRONT
REAR
DOWN

FIG. 4


401
402
FIRST POWER SUPPLY
SECOND POWER SUPPLY
403
POWER SUPPLY SWITCHING CIRCUIT
53
ECU
63
MICROCOMPUTER
71
FIRST DRIVE CIRCUIT
SECOND DRIVE CIRCUIT
72
M
M
13
30
82
83
84
881
882
80
86
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S
812
81
811

FIG. 5

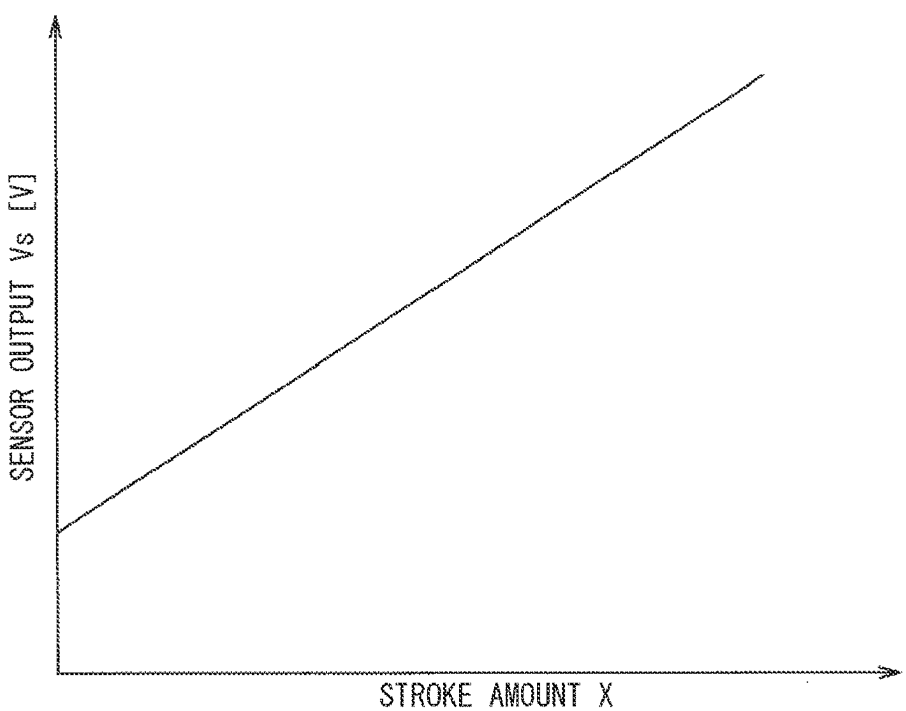
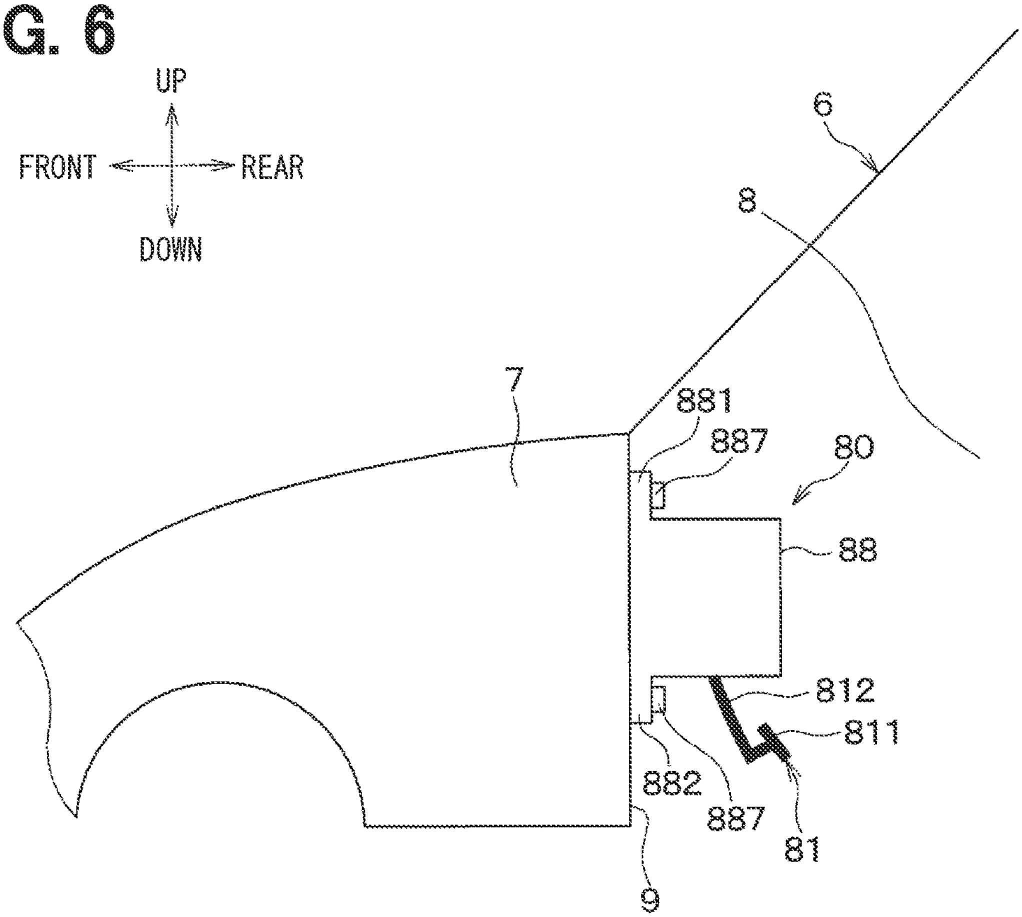

SENSOR OUTPUT Vs [V]
STROKE AMOUNT X

FIG. 6


UP
FRONT
REAR
DOWN
6
8
7
881
887
80
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882
887
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9

FIG. 7

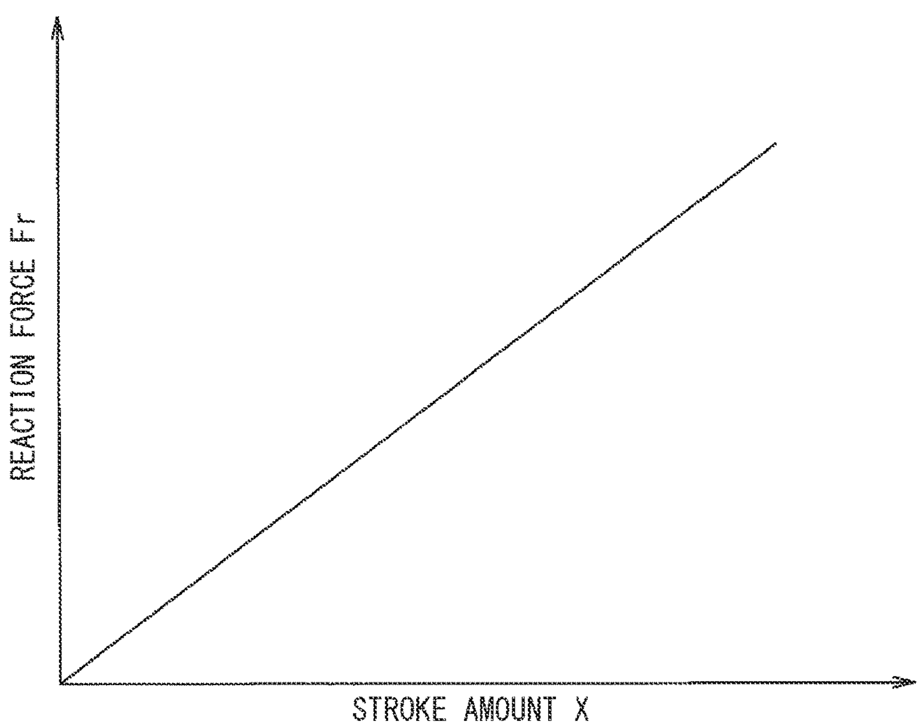

REACTION FORCE Fr
STROKE AMOUNT X

FIG. 8

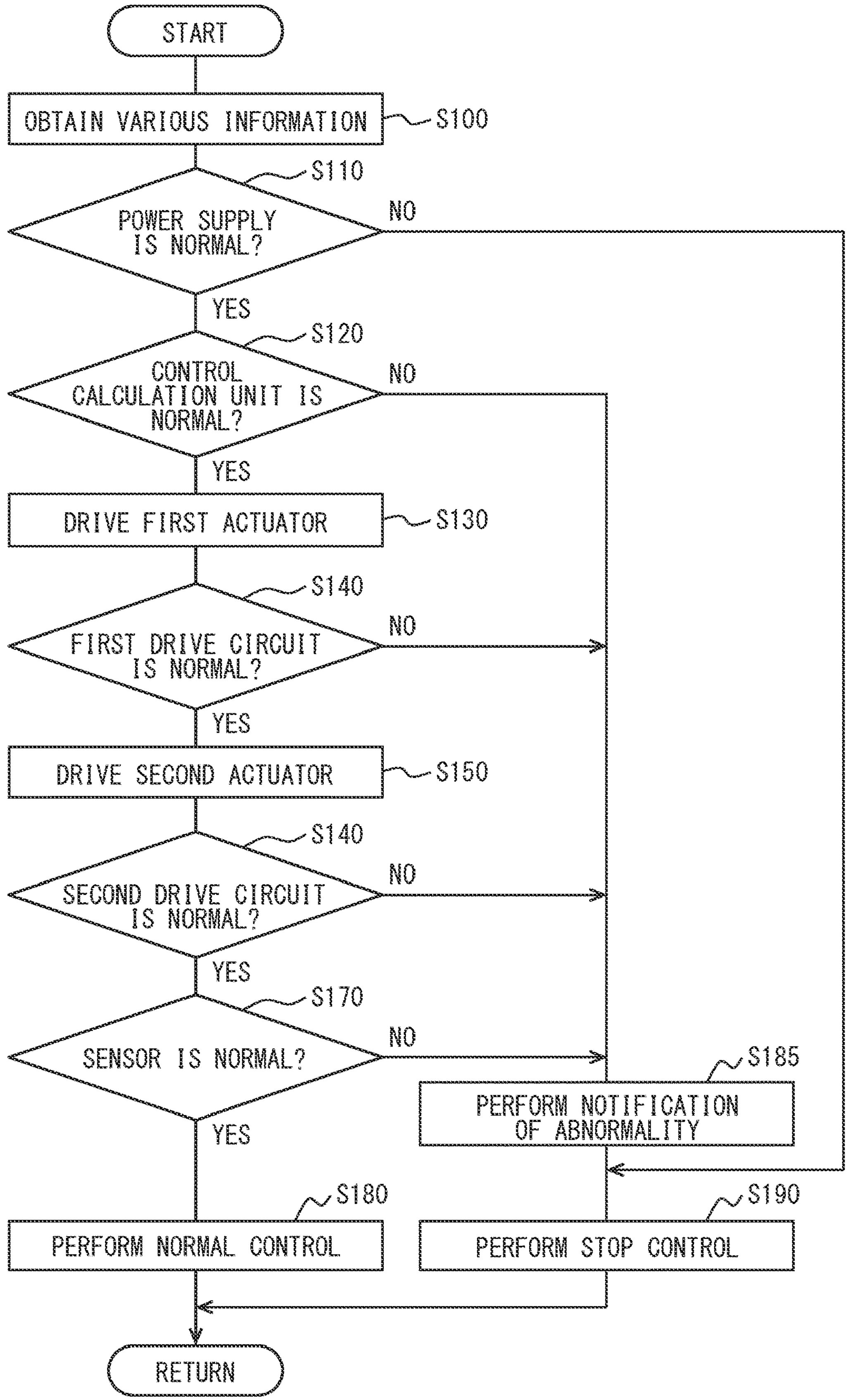

START
OBTAIN VARIOUS INFORMATION
S100
S110
POWER SUPPLY IS NORMAL?
NO
YES
S120
CONTROL CALCULATION UNIT IS NORMAL?
NO
YES
DRIVE FIRST ACTUATOR
S130
S140
FIRST DRIVE CIRCUIT IS NORMAL?
NO
YES
DRIVE SECOND ACTUATOR
S150
S140
SECOND DRIVE CIRCUIT IS NORMAL?
NO
YES
S170
SENSOR IS NORMAL?
NO
YES
S185
PERFORM NOTIFICATION OF ABNORMALITY
S180
PERFORM NORMAL CONTROL
S190
PERFORM STOP CONTROL
RETURN

FIG. 9

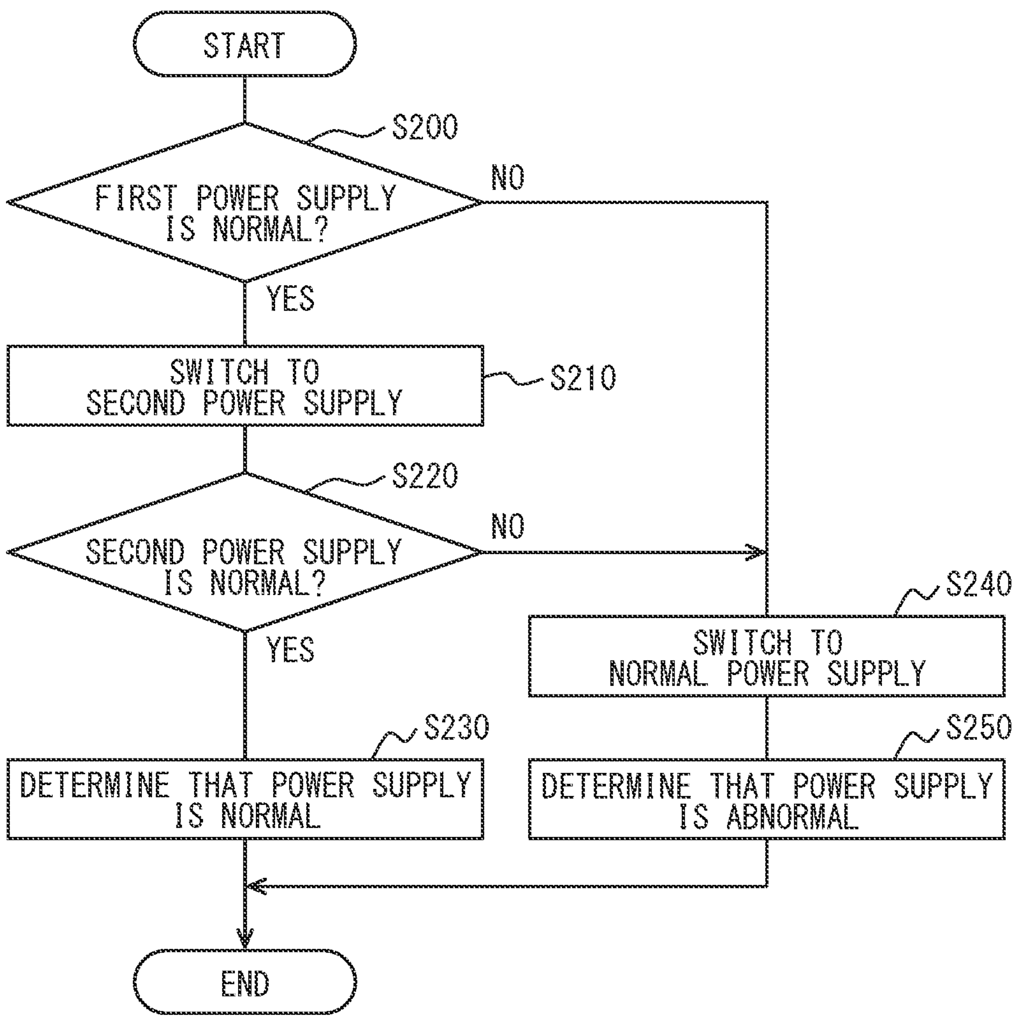
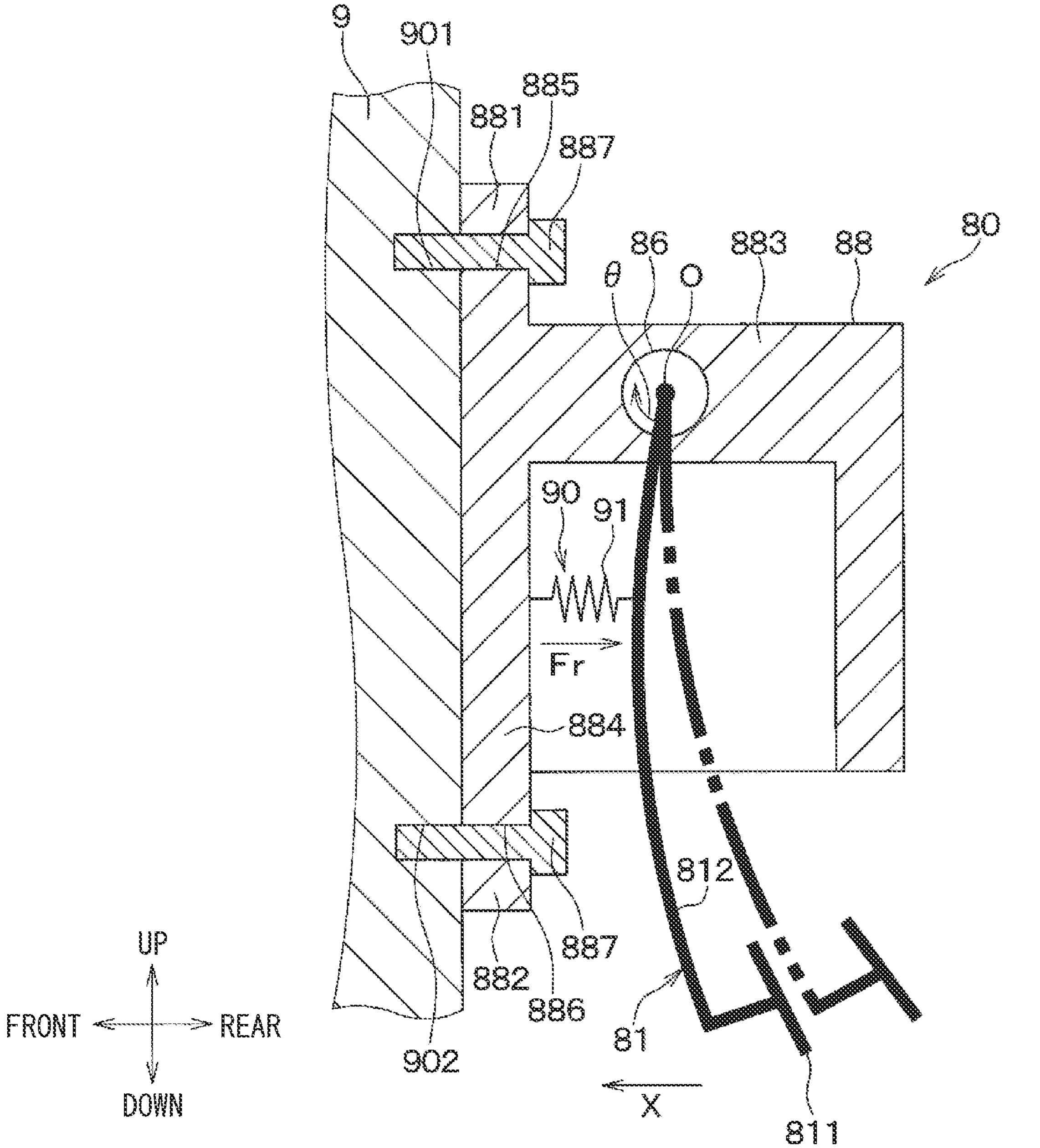

START
S200
FIRST POWER SUPPLY IS NORMAL?
NO
YES
SWITCH TO SECOND POWER SUPPLY
S210
S220
SECOND POWER SUPPLY IS NORMAL?
NO
YES
S240
SWITCH TO NORMAL POWER SUPPLY
S230
DETERMINE THAT POWER SUPPLY IS NORMAL
S250
DETERMINE THAT POWER SUPPLY IS ABNORMAL
END

FIG. 10


9
901
885
881
887
86
883
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80
θ
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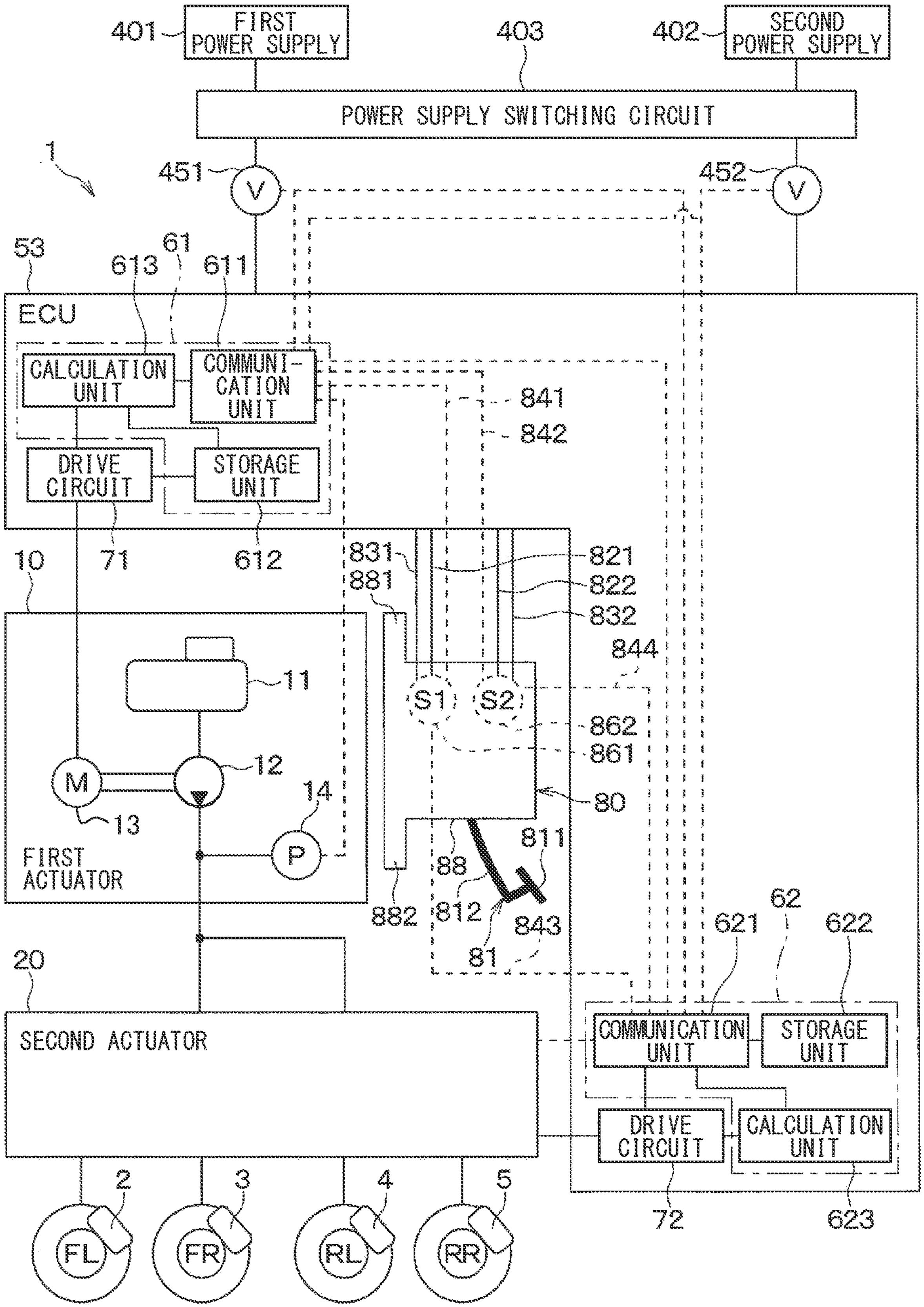
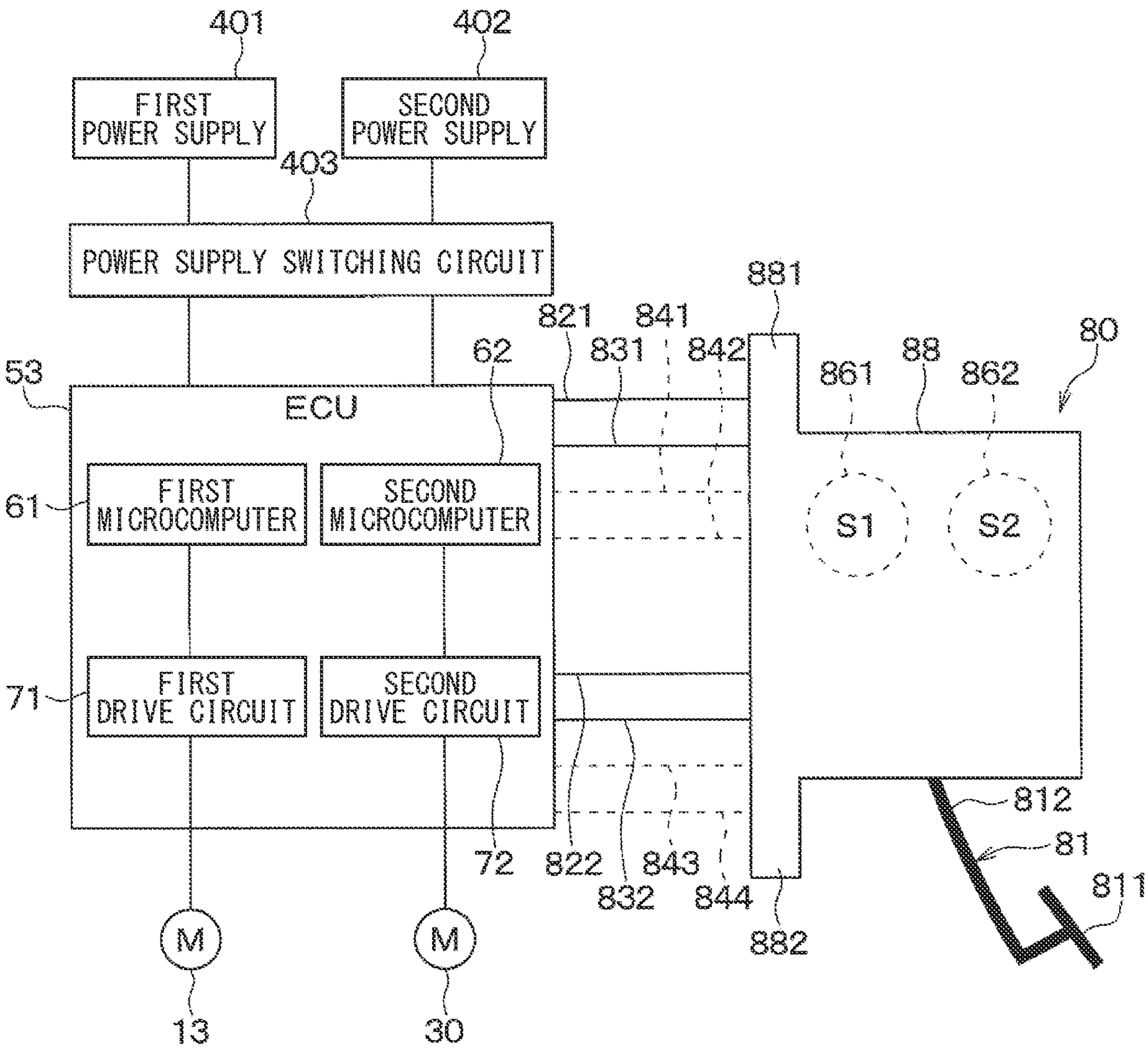
Fr
884
812
887
882
886
902
81
811
X
UP
DOWN
FRONT
REAR

FIG. 11



401
FIRST POWER SUPPLY
403
402
SECOND POWER SUPPLY
POWER SUPPLY SWITCHING CIRCUIT
1
451
V
452
V
53
613
61
611
ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
841
842
DRIVE CIRCUIT
STORAGE UNIT
10
71
612
831
881
821
822
832
844
11
S1
S2
862
861
M
12
14
80
P
13
811
88
FIRST ACTUATOR
882
812
843
62
621
622
81
20
SECOND ACTUATOR
COMMUNICATION UNIT
STORAGE UNIT
DRIVE CIRCUIT
CALCULATION UNIT
2
3
4
5
72
623
FL
FR
RL
RR

FIG. 12



401
402
FIRST POWER SUPPLY
SECOND POWER SUPPLY
403
POWER SUPPLY SWITCHING CIRCUIT
53
ECU
61
FIRST MICROCOMPUTER
62
SECOND MICROCOMPUTER
71
FIRST DRIVE CIRCUIT
SECOND DRIVE CIRCUIT
72
M
M
13
30
821
831
841
842
822
832
843
844
881
882
88
861
862
S1
S2
80
812
81
811

FIG. 13

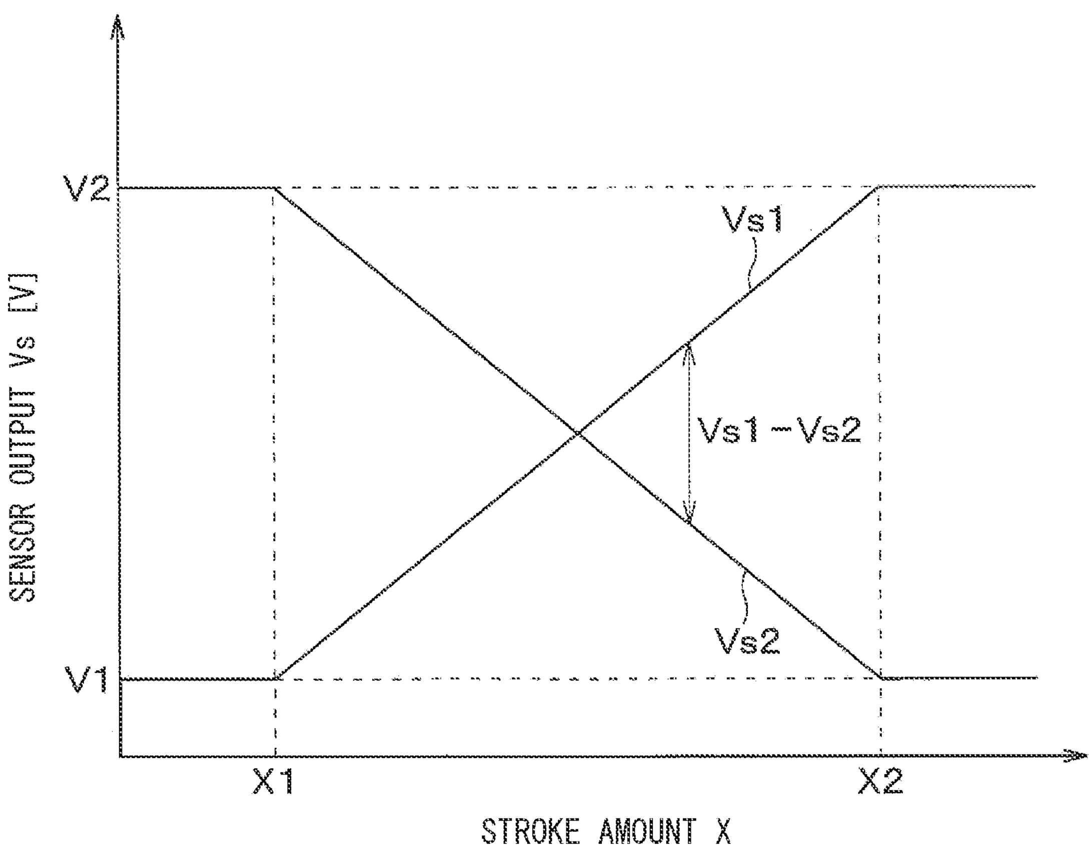


FIG. 14

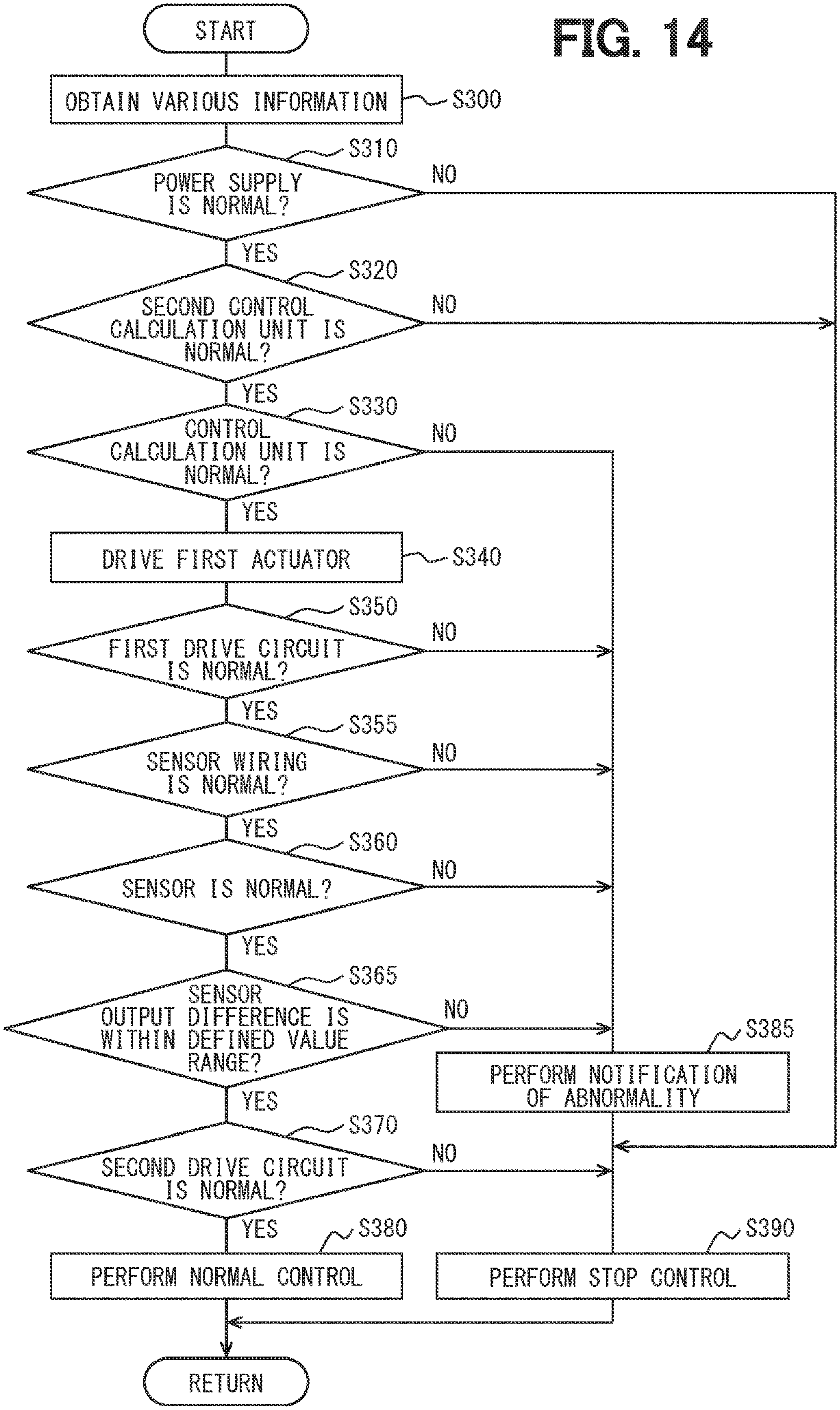

START
OBTAIN VARIOUS INFORMATION
S300
S310
POWER SUPPLY IS NORMAL?
NO
YES
S320
SECOND CONTROL CALCULATION UNIT IS NORMAL?
NO
YES
S330
CONTROL CALCULATION UNIT IS NORMAL?
NO
YES
DRIVE FIRST ACTUATOR
S340
S350
FIRST DRIVE CIRCUIT IS NORMAL?
NO
YES
S355
SENSOR WIRING IS NORMAL?
NO
YES
S360
SENSOR IS NORMAL?
NO
YES
S365
SENSOR OUTPUT DIFFERENCE IS WITHIN DEFINED VALUE RANGE?
NO
S385
PERFORM NOTIFICATION OF ABNORMALITY
YES
S370
SECOND DRIVE CIRCUIT IS NORMAL?
NO
YES
S380
S390
PERFORM NORMAL CONTROL
PERFORM STOP CONTROL
RETURN

FIG. 15

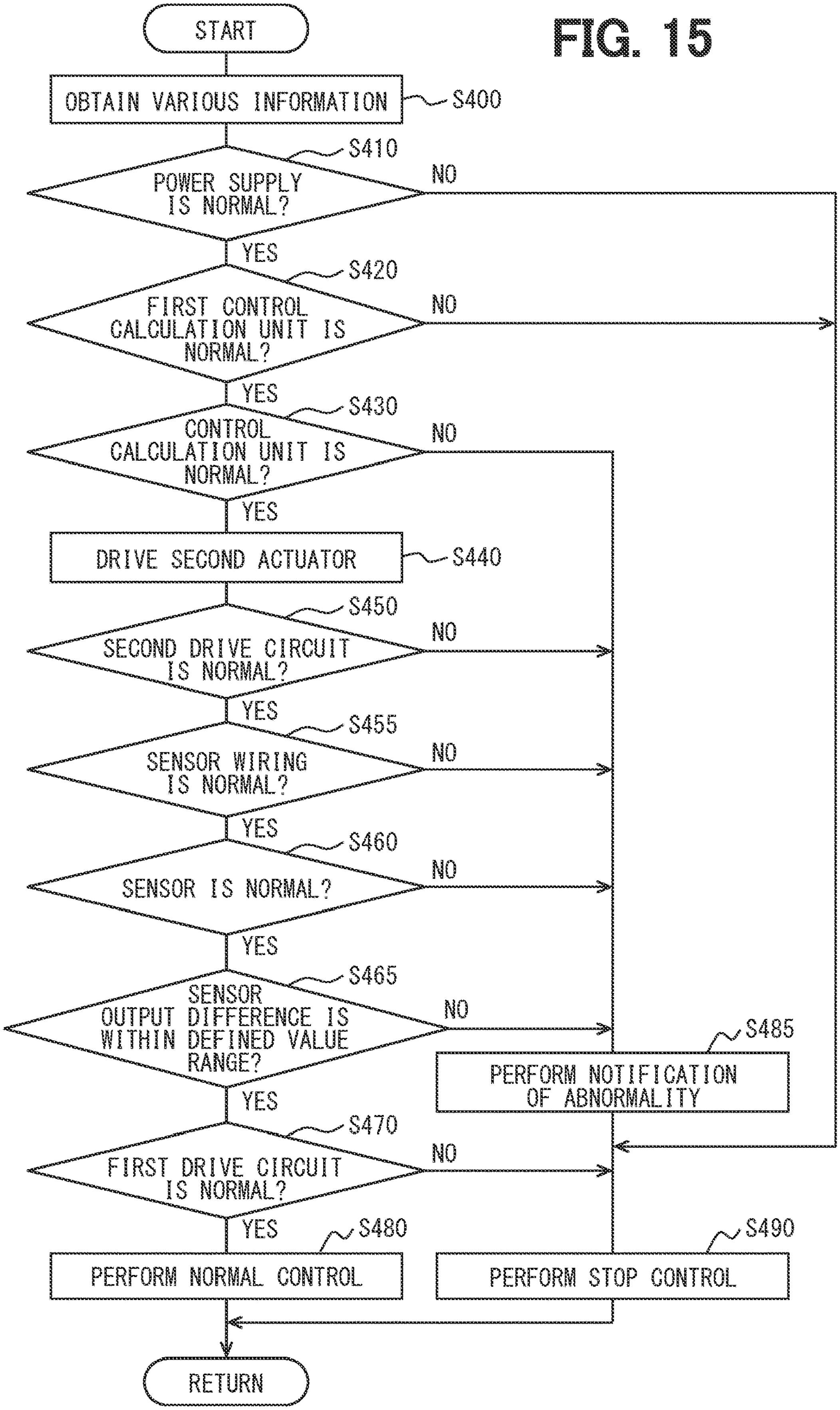
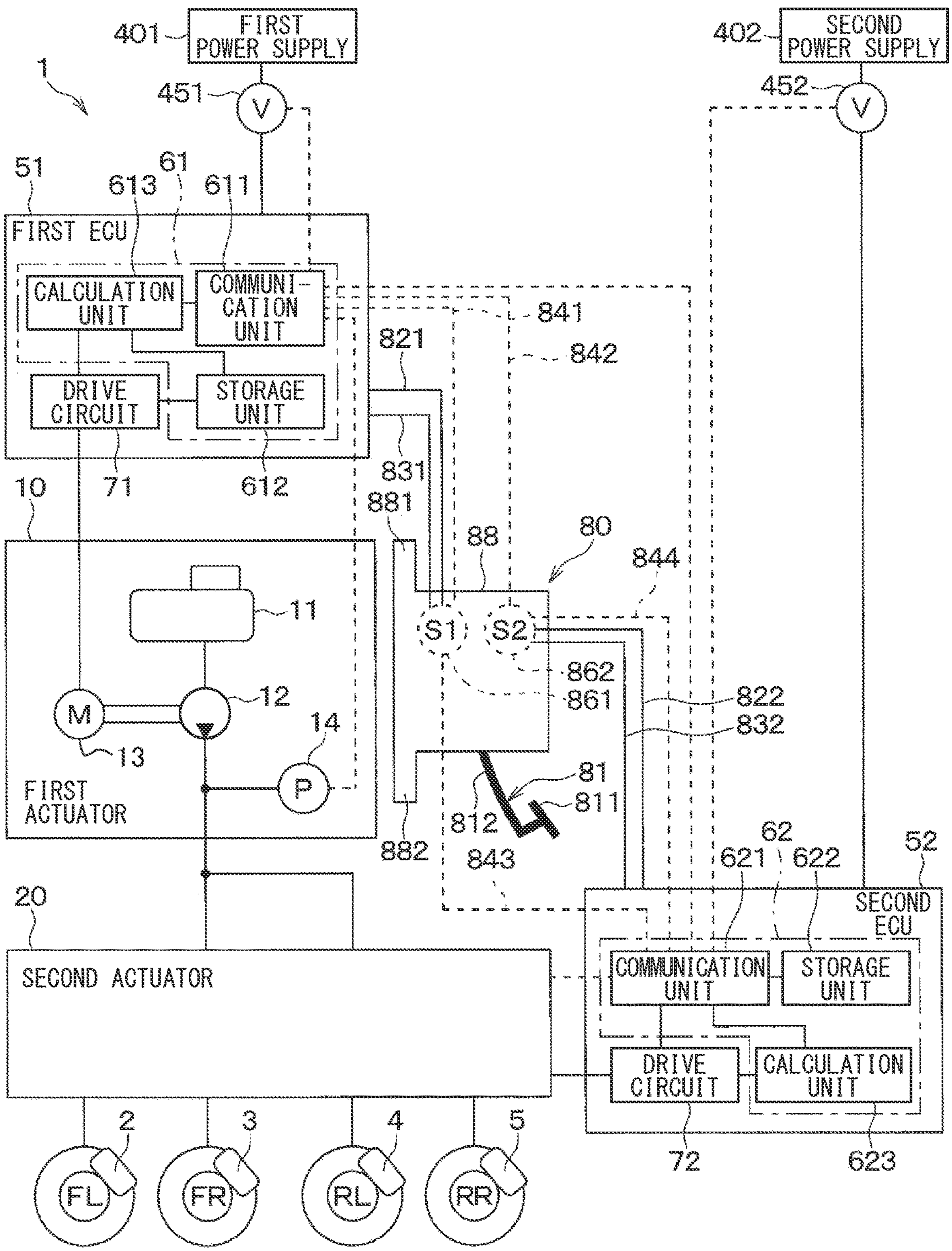

START
OBTAIN VARIOUS INFORMATION
S400
S410
POWER SUPPLY IS NORMAL?
NO
YES
S420
FIRST CONTROL CALCULATION UNIT IS NORMAL?
NO
YES
S430
CONTROL CALCULATION UNIT IS NORMAL?
NO
YES
DRIVE SECOND ACTUATOR
S440
S450
SECOND DRIVE CIRCUIT IS NORMAL?
NO
YES
S455
SENSOR WIRING IS NORMAL?
NO
YES
S460
SENSOR IS NORMAL?
NO
YES
S465
SENSOR OUTPUT DIFFERENCE IS WITHIN DEFINED VALUE RANGE?
NO
S485
PERFORM NOTIFICATION OF ABNORMALITY
YES
S470
FIRST DRIVE CIRCUIT IS NORMAL?
NO
YES
S480
PERFORM NORMAL CONTROL
S490
PERFORM STOP CONTROL
RETURN

FIG. 16


401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
1
451
V
452
V
51
613
61
611
FIRST ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
DRIVE CIRCUIT
STORAGE UNIT
841
842
821
831
10
71
612
881
88
80
844
11
S1
S2
12
862
861
822
832
M
13
14
P
FIRST ACTUATOR
81
811
812
882
843
62
621
622
52
SECOND ECU
20
SECOND ACTUATOR
COMMUNICATION UNIT
STORAGE UNIT
DRIVE CIRCUIT
CALCULATION UNIT
2
3
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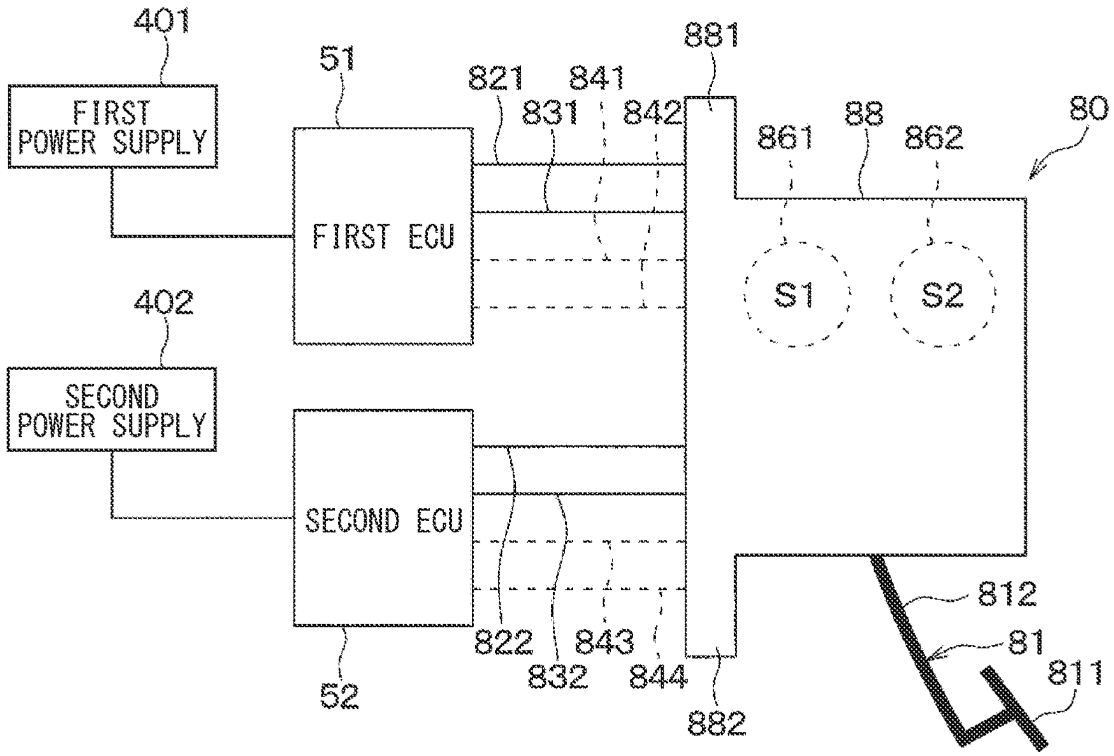
FL
FR
RL
RR
72
623

FIG. 17


401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
51
FIRST ECU
SECOND ECU
52
821
831
841
842
881
861
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862
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S1
S2
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843
844
882
812
81
811

FIG. 18

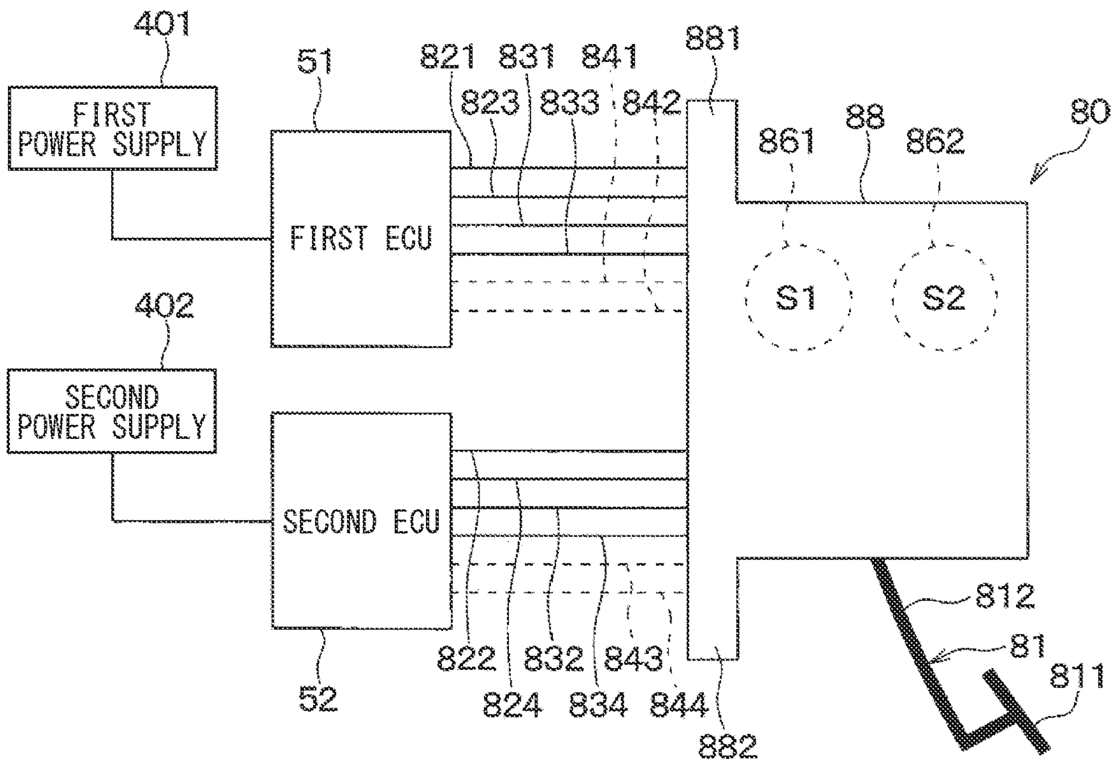
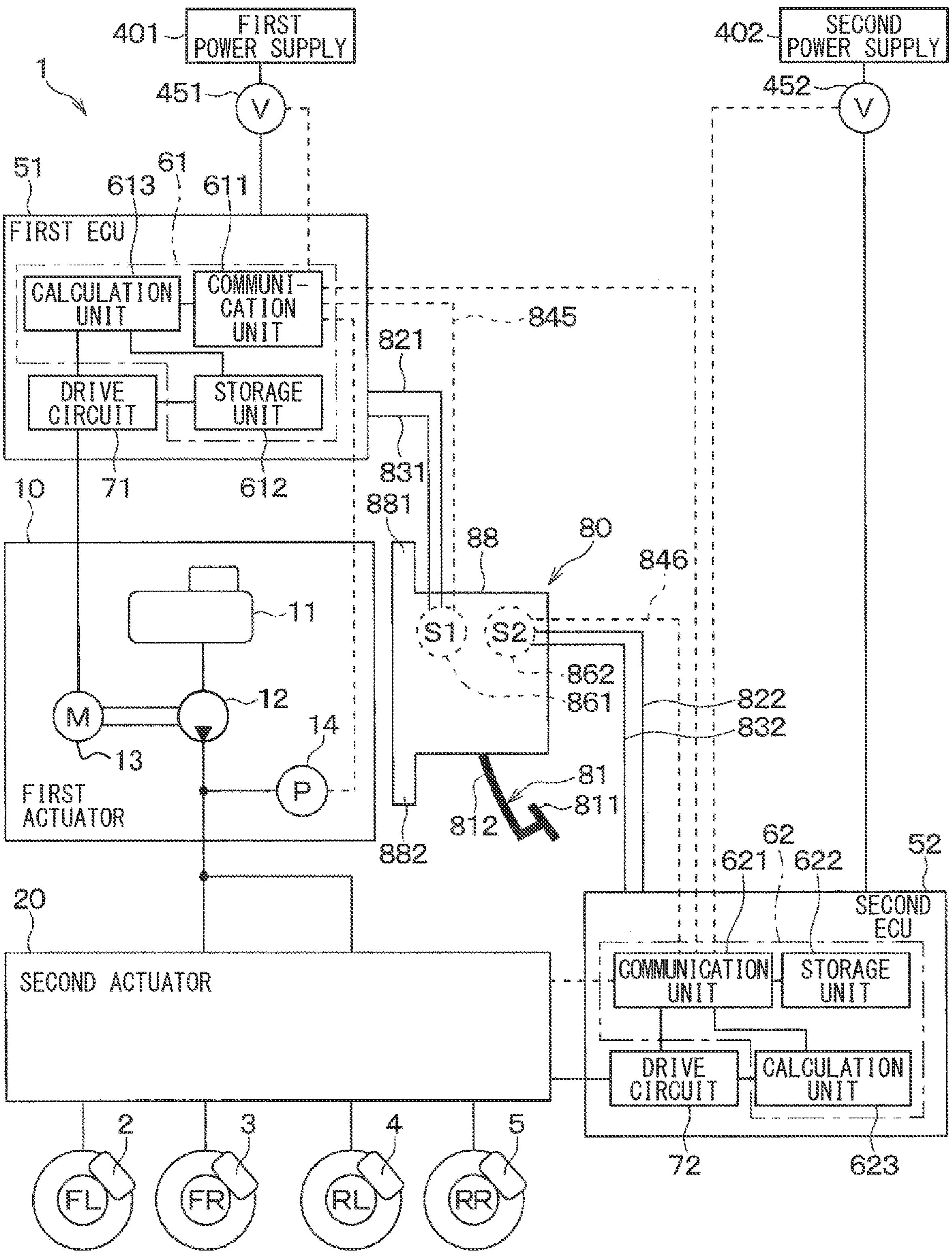

401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
51
FIRST ECU
SECOND ECU
52
821
823
831
833
841
842
881
861
88
862
80
S1
S2
822
824
832
834
843
844
882
812
81
811

FIG. 19


401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
451
V
452
V
1
51
613
61
611
FIRST ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
845
821
DRIVE CIRCUIT
STORAGE UNIT
831
10
71
612
881
88
80
846
11
S1
S2
862
861
822
832
M
12
13
14
P
FIRST ACTUATOR
81
811
812
882
62
621
622
52
20
SECOND ECU
SECOND ACTUATOR
COMMUNICATION UNIT
STORAGE UNIT
DRIVE CIRCUIT
CALCULATION UNIT
72
623
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FL
FR
RL
RR

FIG. 20

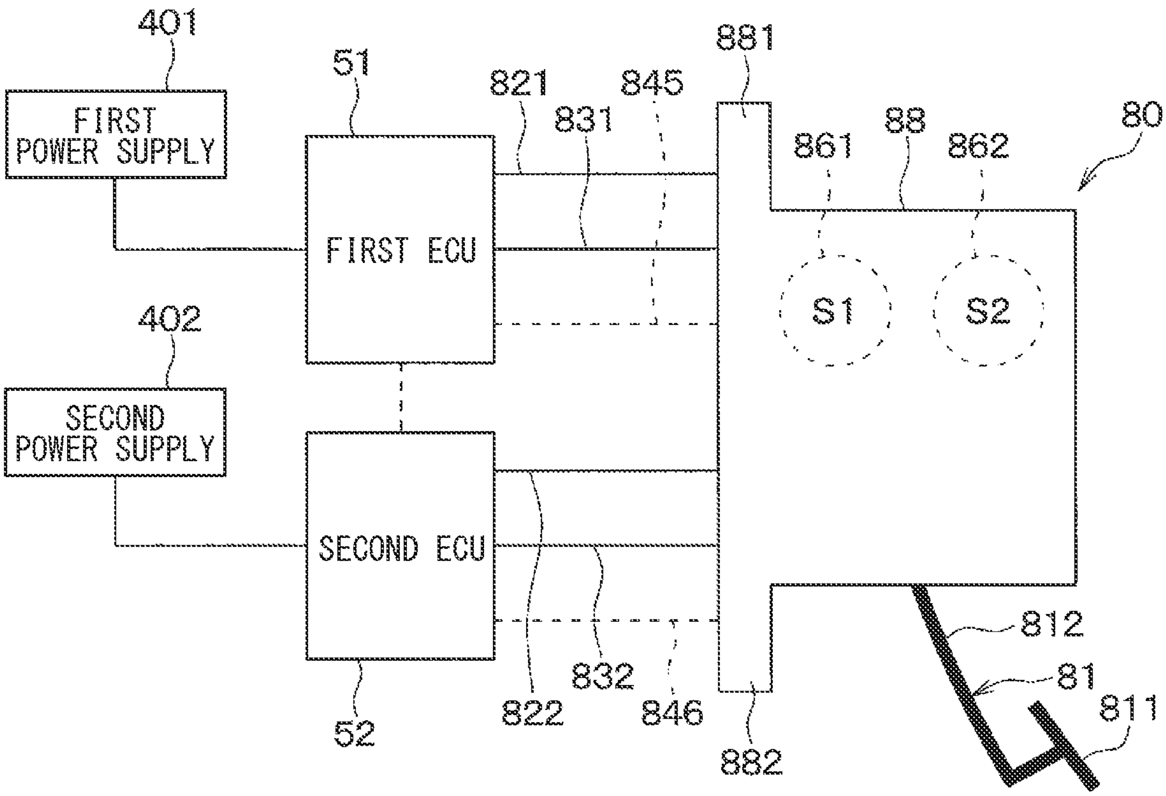
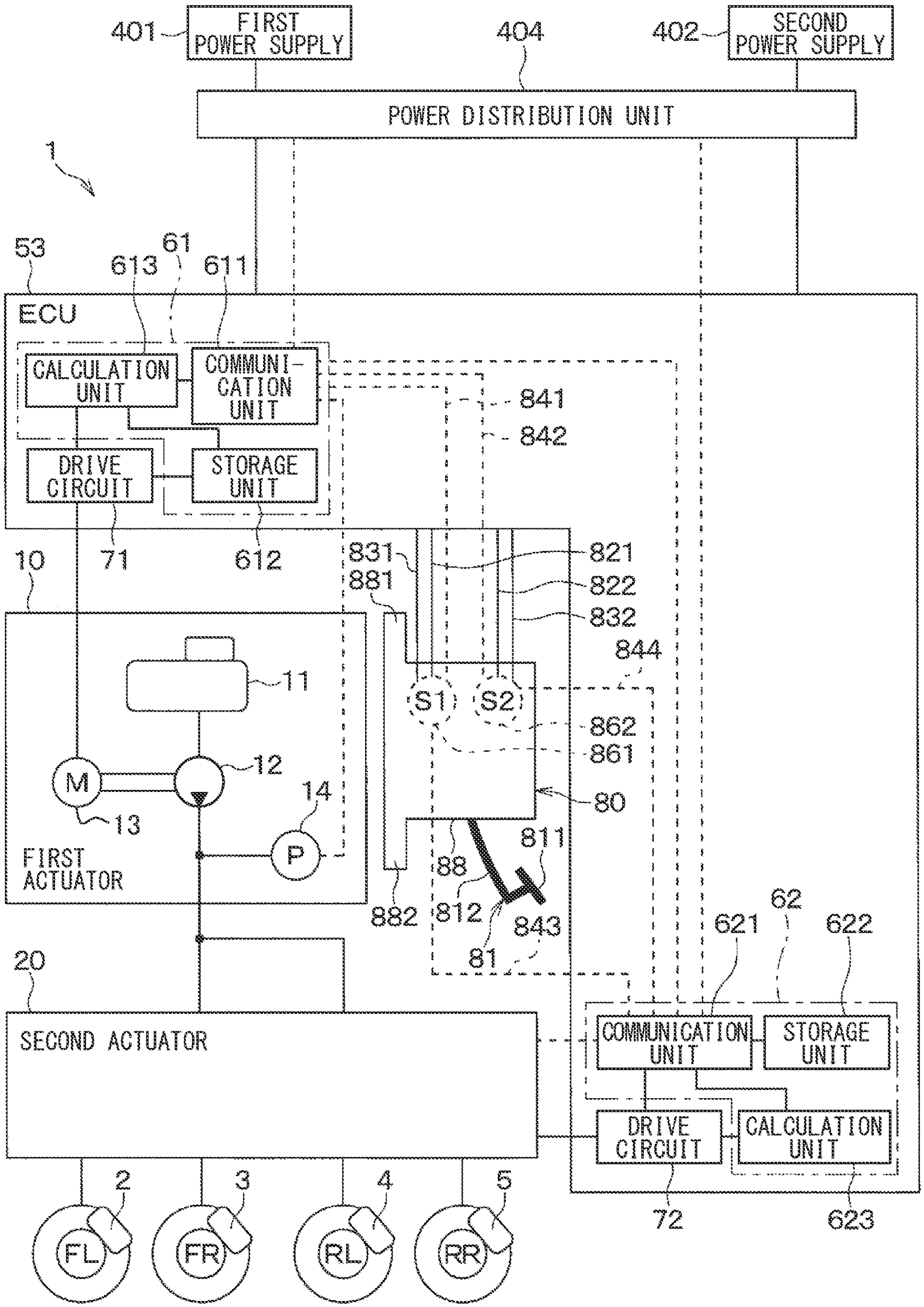

401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
51
FIRST ECU
52
SECOND ECU
821
831
845
822
832
846
881
882
861
88
862
80
S1
S2
812
81
811

FIG. 21



401
FIRST POWER SUPPLY
404
402
SECOND POWER SUPPLY
POWER DISTRIBUTION UNIT
1
53
613
61
611
ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
841
842
DRIVE CIRCUIT
STORAGE UNIT
10
71
612
831
881
821
822
832
844
11
S1
S2
862
861
12
M
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13
P
811
FIRST ACTUATOR
88
882
812
843
81
62
621
622
20
SECOND ACTUATOR
COMMUNICATION UNIT
STORAGE UNIT
DRIVE CIRCUIT
CALCULATION UNIT
2
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FL
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RL
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72
623

FIG. 22

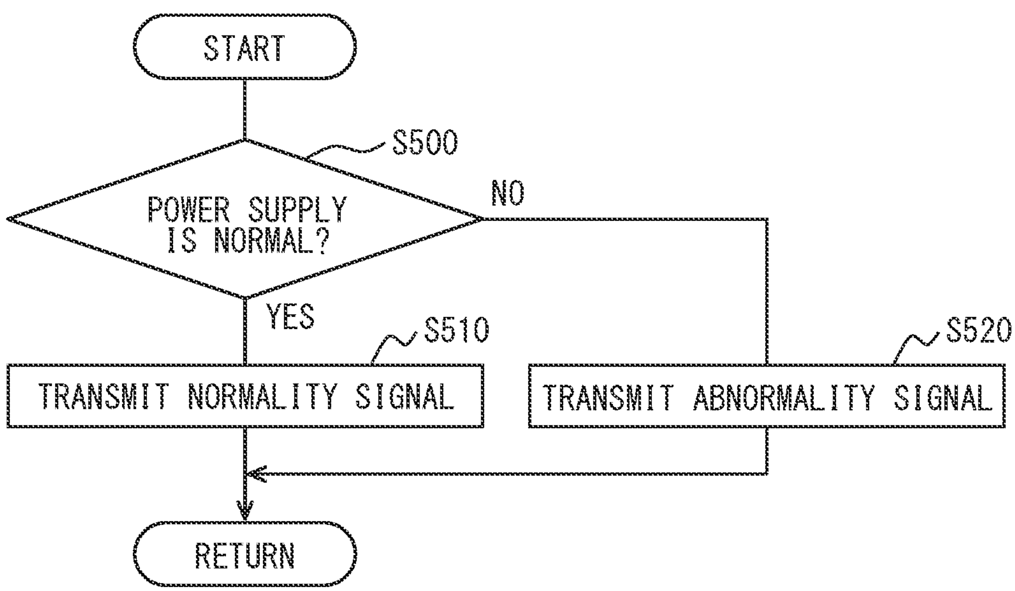
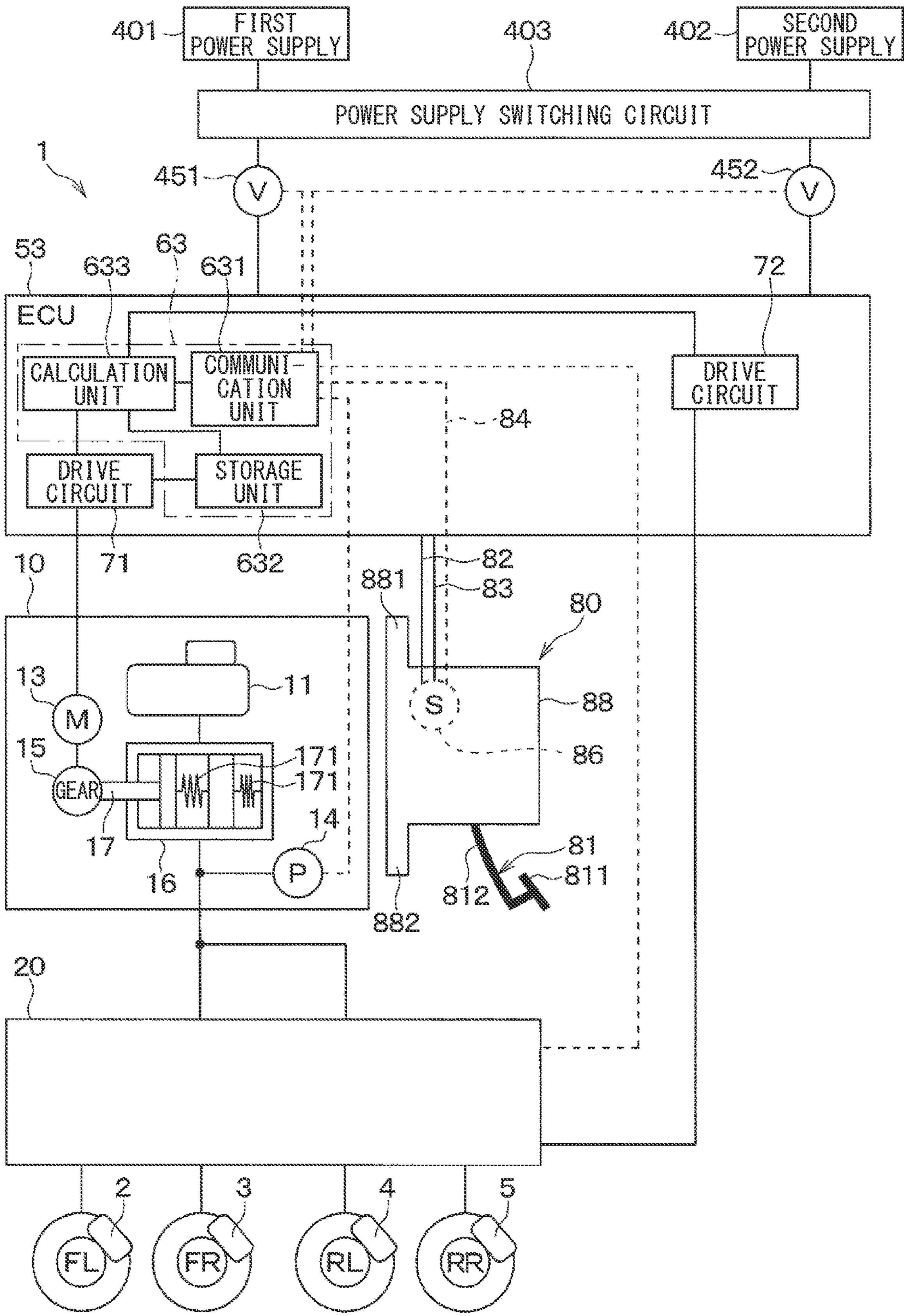

START
S500
POWER SUPPLY IS NORMAL?
NO
YES
S510
TRANSMIT NORMALITY SIGNAL
S520
TRANSMIT ABNORMALITY SIGNAL
RETURN

FIG. 23


401
FIRST POWER SUPPLY
403
402
SECOND POWER SUPPLY
POWER SUPPLY SWITCHING CIRCUIT
1
451
V
452
V
53
633
63
631
72
ECU
CALCULATION UNIT
COMMUNI-CATION UNIT
DRIVE CIRCUIT
84
DRIVE CIRCUIT
STORAGE UNIT
10
71
632
82
83
881
80
13
M
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S
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15
GEAR
171
171
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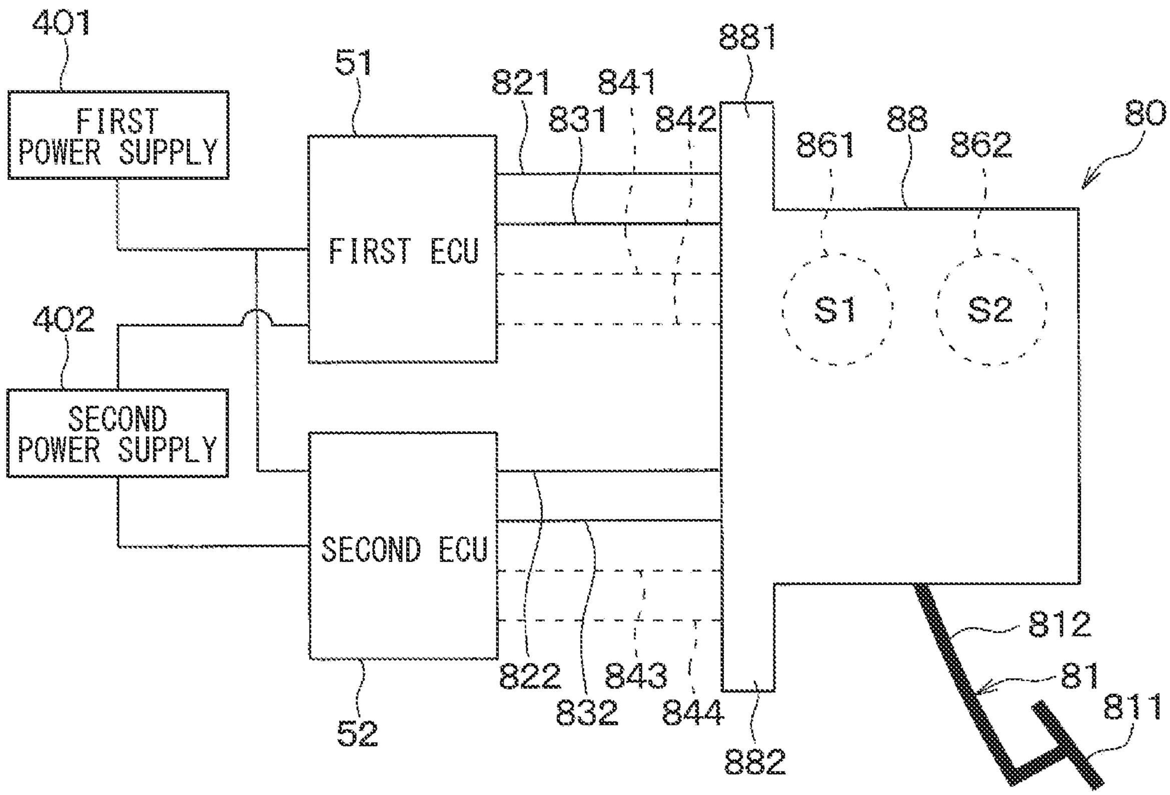
FIG. 24


401
FIRST POWER SUPPLY
402
SECOND POWER SUPPLY
51
FIRST ECU
SECOND ECU
52
821
831
841
842
881
861
88
862
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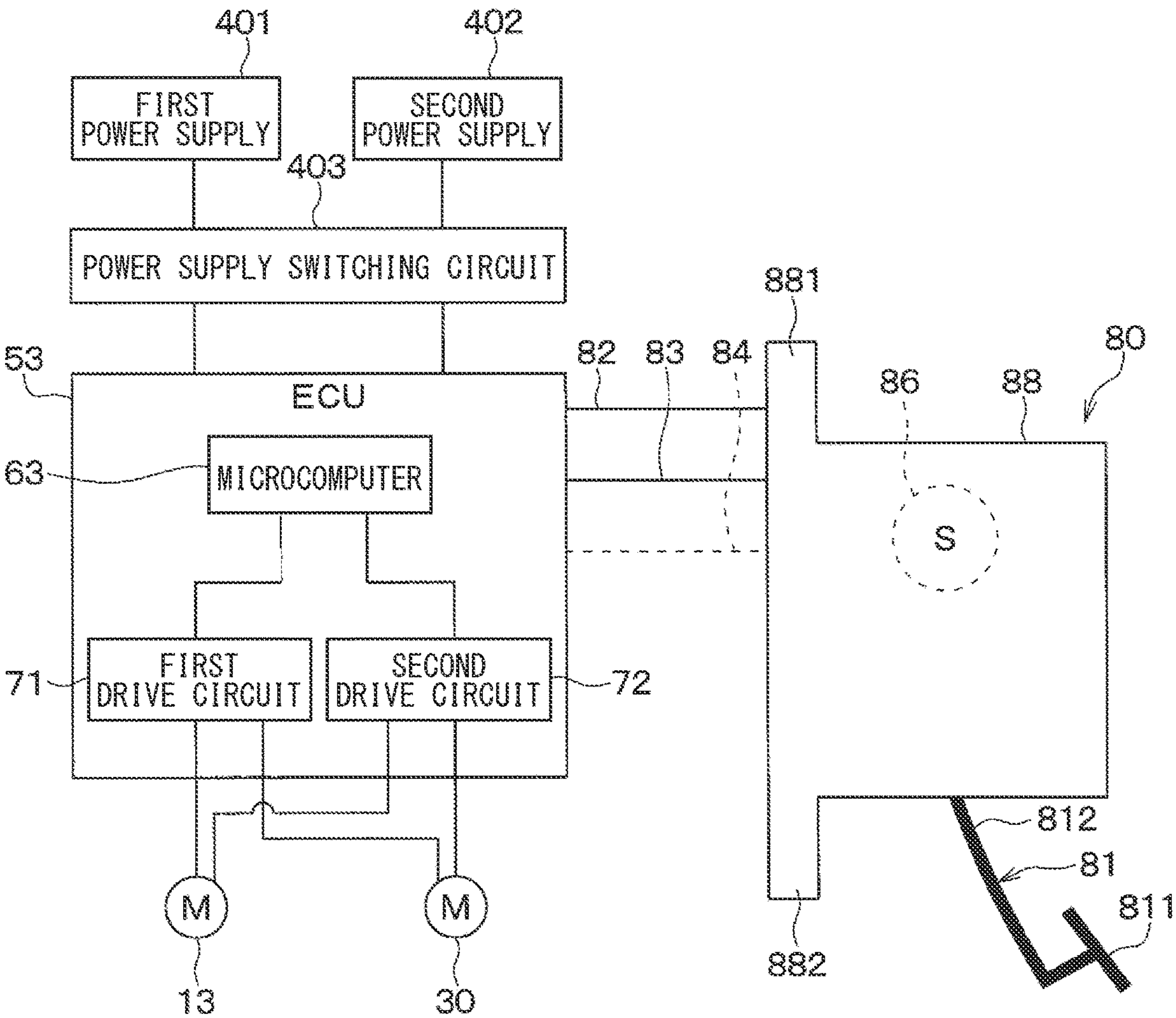
S1
S2
822
832
843
844
882
812
81
811

FIG. 25


401
402
FIRST POWER SUPPLY
SECOND POWER SUPPLY
403
POWER SUPPLY SWITCHING CIRCUIT
881
80
53
82
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84
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88
ECU
63
MICROCOMPUTER
S
71
FIRST DRIVE CIRCUIT
SECOND DRIVE CIRCUIT
72
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VEHICLE BRAKE SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation application of International Patent Application No. PCT/JP2020/046371 filed on Dec. 11, 2020, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2019-225659 filed on Dec. 13, 2019. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a vehicle brake system.

BACKGROUND

A vehicle brake system including a main power supply and a sub power supply has been proposed.

SUMMARY

The present disclosure provides a vehicle brake system. The vehicle brake system includes a vehicle brake device and a hydraulic pressure generation unit. The vehicle brake device includes: a brake pedal that has a pedal portion, and a lever portion rotating about a rotation axis in response to the pedal portion being operated; a stroke sensor that outputs a signal based on a stroke amount of the brake pedal; a housing that rotatably supports the lever portion; and a reaction force generation portion that generates a reaction force applied on the lever portion based on the stroke amount. The hydraulic pressure generation unit generates a hydraulic pressure for braking a vehicle.

BRIEF DESCRIPTION OF DRAWINGS

The features and advantages of the present disclosure will become more apparent from the following detailed description made with reference to the accompanying drawings. In the drawings:

FIG. **1** is a configuration diagram of a vehicle brake system according to a first embodiment;

FIG. **2** is a configuration diagram of a second actuator;

FIG. **3** is a cross-sectional view of a vehicle brake device;

FIG. **4** is a wiring diagram of the vehicle brake system;

FIG. **5** is a diagram illustrating a relationship between stroke amount and sensor output;

FIG. **6** is a view when the vehicle brake device is mounted to a vehicle;

FIG. **7** is a diagram illustrating a relationship between the stroke amount and reaction force;

FIG. **8** is a flowchart illustrating processing executed by a control calculation section;

FIG. **9** is a sub flowchart illustrating processing executed by the control calculation section;

FIG. **10** is a cross-sectional view of the vehicle brake device when a brake pedal is pressed;

FIG. **11** is a configuration diagram of a vehicle brake system according to a second embodiment;

FIG. **12** is a wiring diagram of the vehicle brake system;

FIG. **13** is a diagram illustrating a relationship between the stroke amount and sensor output;

FIG. **14** is a flowchart illustrating processing executed by a first control calculation section;

FIG. **15** is a flowchart illustrating processing executed by a second control calculation section;

FIG. **16** is a configuration diagram of a vehicle brake system according to a third embodiment;

FIG. **17** is a wiring diagram of the vehicle brake system;

FIG. **18** is a wiring diagram of a vehicle brake system according to a fourth embodiment;

FIG. **19** is a configuration diagram of a vehicle brake system according to a fifth embodiment;

FIG. **20** is a wiring diagram of the vehicle brake system;

FIG. **21** is a configuration diagram of a vehicle brake system according to a sixth embodiment;

FIG. **22** is a flowchart of a power distribution unit;

FIG. **23** is a configuration diagram of a vehicle brake system according to a seventh embodiment;

FIG. **24** is a wiring diagram of a vehicle brake system according to an eighth embodiment; and

FIG. **25** is a wiring diagram of a vehicle brake system according to another embodiment.

DETAILED DESCRIPTION

According to the inventors' studies, even when the main power supply and the sub power supply are provided, a vehicle may not be braked if a drive circuit for driving an actuator such as a motor fails. Thus, redundancy of the system may not be ensured in some cases.

The present disclosure provides a vehicle brake system that improves redundancy of the system.

An exemplary embodiment of the present disclosure provides a vehicle brake system that includes a vehicle brake device, a hydraulic pressure generation unit, a hydraulic pressure control device, a first power supply, and a second power supply. The vehicle brake device includes: a brake pedal that has a pedal portion, and a lever portion rotating about a rotation axis in response to the pedal portion being operated; a stroke sensor that outputs a signal based on a stroke amount of the brake pedal; a housing that rotatably supports the lever portion; and a reaction force generation portion that generates a reaction force applied on the lever portion based on the stroke amount. The hydraulic pressure generation unit generates a hydraulic pressure for braking a vehicle. The hydraulic pressure control device includes: a first drive circuit that drives the hydraulic pressure generation unit; and a second drive circuit that drives the hydraulic pressure generation unit. The hydraulic pressure control device controls the first drive circuit and the second drive circuit based on the signal from the stroke sensor to control the hydraulic pressure generated by the hydraulic pressure generation unit. The first power supply supplies power to the hydraulic pressure control device. The second power supply supplies power to the hydraulic pressure control device.

In the exemplary embodiment of the present disclosure, the redundancy of the vehicle brake system can be improved.

Hereinafter, embodiments will be described with reference to the drawings. In the following embodiments, the same or equivalent portions are denoted by the same reference signs, and the description thereof will be omitted.

First Embodiment

A vehicle brake system **1** according to a first embodiment controls a left front wheel FL, a right front wheel FR, a left rear wheel RL, and a right rear wheel RR, which are the wheels of a vehicle **6**. First, the vehicle brake system **1** will be described.

As illustrated in FIG. **1**, the vehicle brake system **1** includes a wheel cylinder for the left front wheel, a wheel cylinder for the right front wheel, a wheel cylinder for the left rear wheel, and a wheel cylinder for the right rear wheel. The vehicle brake system **1** further includes a first actuator **10**, a second actuator **20**, a first power supply **401**, a first voltage sensor **451**, a second power supply **402**, a second voltage sensor **452**, a power supply switching circuit **403**, an ECU **53**, and a vehicle brake device **80**. Hereinafter, each of the wheel cylinders is referred to as the W/C for convenience. The ECU is an abbreviation for Electronic Control Unit.

The left front wheel W/C **2** is disposed on the left front wheel FL. The right front wheel W/C **3** is disposed on the right front wheel FR. The left rear wheel W/C **4** is disposed on the left rear wheel RL. The right rear wheel W/C **5** is disposed on the right rear wheel RR. The left front wheel W/C **2**, the right front wheel W/C **3**, the left rear wheel W/C **4**, and the right rear wheel W/C **5** are connected to respective brake pads (not illustrated) of the vehicle **6**.

The first actuator **10** corresponds to a first hydraulic pressure generation unit, and generates a brake hydraulic pressure. The first actuator **10** increases the brake hydraulic pressures to cause respective brake hydraulic pressures of the left front wheel W/C **2**, the right front wheel W/C **3**, the left rear wheel W/C **4**, and the right rear wheel W/C **5** to increase. Specifically, the first actuator **10** includes a reservoir **11**, a first pump **12**, a first actuator motor **13**, and a first pressure sensor **14**.

The reservoir **11** stores a brake fluid such as oil or the like, and also supplies the brake fluid to the first pump **12**.

The first pump **12** is driven by the first actuator motor **13**, which corresponds to a first motor. With this driving, the first pump **12** increases a pressure of the brake fluid from the reservoir **11**. The brake fluid with the increased hydraulic pressure flows from the first actuator **10** to the second actuator **20**.

The first pressure sensor **14** outputs, to the ECU **53**, a signal being in accordance with the hydraulic pressure of the brake fluid flowing to the second actuator **20**.

The second actuator **20** corresponds to a second hydraulic pressure generation unit, and generates brake hydraulic pressures. The second actuator **20** controls the respective brake hydraulic pressures of the left front wheel W/C **2**, the right front wheel W/C **3**, the left rear wheel W/C **4**, and the right rear wheel W/C **5**, based on signals from the ECU **53** described later. For example, as illustrated in FIG. **2**, the second actuator **20** includes a first piping system **21**, a second piping system **26**, and a second actuator motor **30**.

The first piping system **21** controls the brake hydraulic pressures of the left front wheel W/C **2** and the right front wheel W/C **3**. Specifically, the first piping system **21** includes a first main pipe line **211**, a first differential pressure control valve **212**, a second pressure sensor **213**, a first branch pipe line **214**, a first pressure increase control valve **215**, and a first pressure reduction control valve **216**. The first piping system **21** also includes a second branch pipe line **217**, a second pressure increase control valve **218**, a second pressure reduction control valve **219**, a first pressure reduction pipe line **220**, a first pressure adjustment reservoir **221**, a first auxiliary pipe line **222**, a first return-flow pipe line **223**, and a second pump **224**.

The first main pipe line **211** is connected to the first actuator **10**, and transmits the brake hydraulic pressure from the first actuator **10** to the first differential pressure control valve **212**.

The first differential pressure control valve **212** controls a differential pressure between an upstream side and a downstream side of the first main pipe line **211** using a signal from the ECU **53** described later. For example, when the brake hydraulic pressures on sides of the left front wheel W/C **2** and the right front wheel W/C **3** are higher than the brake hydraulic pressure on the side of the first actuator **10** by a predetermined pressure or higher, the first differential pressure control valve **212** allows a flow of the brake fluid from the sides of the left front wheel W/C **2** and the right front wheel W/C **3** to the side of the first actuator **10**. As a result, the brake hydraulic pressures on the sides of the left front wheel W/C **2** and the right front wheel W/C **3** are maintained so as not to be higher than the brake hydraulic pressure on the side of the first actuator **10** by a predetermined pressure or higher.

The second pressure sensor **213** outputs a signal being in accordance with the brake hydraulic pressure on the downstream side of the first differential pressure control valve **212** to the ECU **53** described later.

The first branch pipe line **214** directs the brake fluid flowing from the first differential pressure control valve **212** to the first pressure increase control valve **215**.

The first pressure increase control valve **215** is a normally open type two-position electromagnetic valve capable of controlling a communicating state and a blocking state. Specifically, when a solenoid coil (not illustrated) of the first pressure increase control valve **215** is in a non-energized state, the first pressure increase control valve **215** is in the communicating state, and thus allows a flow of the brake fluid to the left front wheel W/C **2** and the first pressure reduction control valve **216**. When the solenoid coil (not illustrated) of the first pressure increase control valve **215** is in an energized state, the first pressure increase control valve **215** becomes the blocking state, and thus blocks the flow of the brake fluid to the left front wheel W/C **2** and the first pressure reduction control valve **216**.

The first pressure reduction control valve **216** is a normally closed type two-position electromagnetic valve capable of controlling a blocking state and a communicating state. Specifically, when a solenoid coil (not illustrated) of the first pressure reduction control valve **216** is in a non-energized state, the first pressure reduction control valve **216** is in the blocking state, and thus blocks a flow of the brake fluid to the first pressure reduction pipe line **220** described later. When the solenoid coil (not illustrated) of the first pressure reduction control valve **216** is in an energized state, the first pressure reduction control valve **216** becomes the communicating state, and thus allows the flow of the brake fluid to the first pressure reduction pipe line **220** described later.

The second branch pipe line **217** directs the brake fluid flowing from the first differential pressure control valve **212** to the second pressure increase control valve **218**.

Similar to the first pressure increase control valve **215**, the second pressure increase control valve **218** is a normally open type two-position electromagnetic valve. Specifically, when a solenoid coil (not illustrated) of the second pressure increase control valve **218** is in a non-energized state, the second pressure increase control valve **218** is in a communicating state, and thus allows a flow of the brake fluid to the right front wheel W/C **3** and the second pressure reduction control valve **219**. When the solenoid coil (not illustrated) of the second pressure increase control valve **218** is in an energized state, the second pressure increase control valve **218** becomes a blocking state, and thus blocks the flow of the brake fluid to the right front wheel W/C **3** and the second pressure reduction control valve **219**.

Similar to the first pressure reduction control valve **216**, the second pressure reduction control valve **219** is a normally closed type two-position electromagnetic valve. Specifically, when a solenoid coil (not illustrated) of the second pressure reduction control valve **219** is in a non-energized state, the second pressure reduction control valve **219** is in the blocking state, and thus blocks a flow of the brake fluid to the first pressure reduction pipe line **220** described later. When the solenoid coil (not illustrated) of the second pressure reduction control valve **219** is in an energized state, the second pressure reduction control valve **219** becomes the communicating state, and thus allows the flow of the brake fluid to the first pressure reduction pipe line **220** described later.

The first pressure reduction pipe line **220** directs the brake fluids flowing from the first pressure reduction control valve **216** and the second pressure reduction control valve **219** to the first pressure adjustment reservoir **221**.

The first auxiliary pipe line **222** branches from the first main pipe line **211**, and directs the brake fluid flowing from the first actuator **10** to the first pressure adjustment reservoir **221**.

The first pressure adjustment reservoir **221** stores the brake fluids flowing from the first pressure reduction control valve **216** and the second pressure reduction control valve **219** through the first pressure reduction pipe line **220**. The first pressure adjustment reservoir **221** also stores the brake fluid flowing from the first actuator **10** through the first auxiliary pipe line **222**. Further, when the brake fluid is sucked by the second pump **224** described later, the first pressure adjustment reservoir **221** adjusts a flow rate of the stored brake fluid.

The first return-flow pipe line **223** is connected between the first differential pressure control valve **212** and each of the first pressure increase control valve **215** and the second pressure increase control valve **218**. The first return-flow pipe line **223** is connected to the second pump **224**.

The second pump **224** is connected to the first pressure reduction pipe line **220**, and is driven by the second actuator motor **30**, which corresponds to a second motor. With this driving, the second pump **224** sucks the brake fluid stored in the first pressure adjustment reservoir **221**. The sucked brake fluid flows between the first differential pressure control valve **212** and each of the first pressure increase control valve **215** and the second pressure increase control valve **218** after the brake fluid flows through the first return-flow pipe line **223**. As a result, the respective brake hydraulic pressures of the left front wheel W/C **2** and the right front wheel W/C **3** increase.

The second piping system **26** controls the brake hydraulic pressures of the left rear wheel W/C **4** and the right rear wheel W/C **5**. Specifically, the second piping system **26** includes a second main pipe line **261**, a second differential pressure control valve **262**, a third pressure sensor **263**, a third branch pipe line **264**, a third pressure increase control valve **265**, and a third pressure reduction control valve **266**. The second piping system **26** also includes a fourth branch pipe line **267**, a fourth pressure increase control valve **268**, a fourth pressure reduction control valve **269**, a second pressure reduction pipe line **270**, a second pressure adjustment reservoir **271**, a second auxiliary pipe line **272**, a second return-flow pipe line **273**, and a third pump **274**.

Herein, the second piping system **26** is configured similarly to the first piping system **21**. Thus, the left front wheel W/C **2** described above is replaced by the right rear wheel W/C **5**. The right front wheel W/C **3** described above is replaced by the left rear wheel W/C **4**. Further, the second main pipe line **261** corresponds to the first main pipe line **211**. The second differential pressure control valve **262** corresponds to the first differential pressure control valve **212**. The third pressure sensor **263** corresponds to the second pressure sensor **213**. The third branch pipe line **264** corresponds to the first branch pipe line **214**. The third pressure increase control valve **265** corresponds to the first pressure increase control valve **215**. The third pressure reduction control valve **266** corresponds to the first pressure reduction control valve **216**. The fourth branch pipe line **267** corresponds to the second branch pipe line **217**. The fourth pressure increase control valve **268** corresponds to the second pressure increase control valve **218**. The fourth pressure reduction control valve **269** corresponds to the second pressure reduction control valve **219**. The second pressure reduction pipe line **270** corresponds to the first pressure reduction pipe line **220**. The second pressure adjustment reservoir **271** corresponds to the first pressure adjustment reservoir **221**. The second auxiliary pipe line **272** corresponds to the first auxiliary pipe line **222**. The second return-flow pipe line **273** corresponds to the first return-flow pipe line **223**. The third pump **274** corresponds to the second pump **224**.

The first power supply **401** supplies power to the ECU **53**.

The first voltage sensor **451** outputs, to the ECU **53**, a signal being in accordance with a voltage applied from the first power supply **401** to the ECU **53**.

The second power supply **402** supplies power to the ECU **53**.

The second voltage sensor **452** outputs, to the ECU **53**, a signal being in accordance with a voltage applied from the second power supply **402** to the ECU **53**.

The power supply switching circuit **403** switches a power supply source that supplies power to the ECU **53** to either the first power supply **401** or the second power supply **402** based on a signal from the ECU **53** described later.

The ECU **53** corresponds to a hydraulic pressure control device, and controls the first actuator **10** by controlling the first actuator motor **13**. The ECU **53** also controls the second actuator **20** by controlling the second actuator motor **30**. Further, the ECU **53** switches the power supply source that supplies power to the ECU **53** to either the first power supply **401** or the second power supply **402** by controlling the power supply switching circuit **403**. Specifically, the ECU **53** includes a microcomputer **63**, a first drive circuit **71**, and a second drive circuit **72**.

The microcomputer **63** corresponds to a hydraulic pressure control unit, and controls the first actuator **10** by controlling the first drive circuit **71** described later. The microcomputer **63** controls the second actuator **20** by controlling the second drive circuit **72** described later. Specifically, the microcomputer **63** includes a communication section **631**, a storage section **632**, and a control calculation section **633**.

The communication section **631** includes an interface for communicating with the first pressure sensor **14**, an interface for communicating with the second pressure sensor **213**, and an interface for communicating with the third pressure sensor **263**. The communication section **631** also includes an interface for communicating with the first voltage sensor **451** and an interface for communicating with the second voltage sensor **452**. The communication section **631** further includes an interface for communicating with a stroke sensor **86** of the vehicle brake device **80**, described later.

The storage section **632** includes non-volatile memories such as a read-only memory (ROM) and a flash memory, and a volatile memory such as a random access memory (RAM). The non-volatile memories and the volatile memory are non-transitory tangible storage media.

The control calculation section **633** includes a central processing unit (CPU) and the like. The control calculation section **633** outputs a signal for driving the first actuator motor **13** to the first drive circuit **71** by executing a program stored in the ROM of the storage section **632**. The control calculation section **633** also outputs a signal for driving the second actuator motor **30** to the second drive circuit **72** by executing a program stored in the ROM of the storage section **632**. The control calculation section **633** further outputs, to the power supply switching circuit **403**, the signal for switching the power supply source that supplies power to the ECU **53** by executing a program stored in the ROM of the storage section **632**.

The first drive circuit **71** includes, for example, a switching element and the like, and drives the first actuator **10** by supplying power to the first actuator motor **13** based on the signal from the control calculation section **633**.

The second drive circuit **72** includes, for example, a switching element and the like, and drives the second actuator **20** by supplying power to each valve of the second actuator **20** and the second actuator motor **30** based on the signal from the control calculation section **633**.

As illustrated in FIGS. **1**, **3**, and **4**, the vehicle brake device **80** includes a brake pedal **81**, a sensor power supply line **82**, a sensor ground line **83**, a sensor output line **84**, the stroke sensor **86**, a housing **88**, and a reaction force generation portion **90**.

The brake pedal **81** is operated by a driver of the vehicle **6** pressing the brake pedal **81**. Specifically, the brake pedal **81** includes a pedal portion **811** and a lever portion **812**. The pedal portion **811** is pressed by the driver of the vehicle **6**. The lever portion **812** is connected to the pedal portion **811**, and rotates about a rotation axis O when the pedal portion **811** is pressed by the driver of the vehicle **6**.

As illustrated in FIGS. **1** and **4**, the sensor power supply line **82** is connected to the ECU **53** and the stroke sensor **86** described later. With this connection, power from the first power supply **401** and the second power supply **402** is supplied to the stroke sensor **86** through the ECU **53** and the sensor power supply line **82**.

The sensor ground line **83** is connected to the ECU **53** and the stroke sensor **86**.

The sensor output line **84** is connected to the ECU **53** and the stroke sensor **86**.

As illustrated in FIG. **3**, the stroke sensor **86** is disposed, for example, next to the rotation axis O of the lever portion **812**. As illustrated in FIGS. **1** and **4**, the stroke sensor **86** outputs a signal being in accordance with a stroke amount X to the ECU **53** through the sensor output line **84**. The stroke amount X is an operation amount of the brake pedal **81** generated by a pedal force of the driver of the vehicle **6**. Herein, the stroke amount X is, for example, an amount of translational movement of the pedal portion **811** toward the front of the vehicle **6**. As illustrated in FIG. **5**, the stroke amount X and a sensor output Vs of the stroke sensor **86** are adjusted to have a linear relationship. Herein, the sensor output Vs is expressed by, for example, voltage. Alternatively, the stroke sensor **86** may also output a signal being in accordance with a rotation angle θ about the rotation axis O of the lever portion **812** to the ECU **53** through the sensor output line **84**. At this time, similarly to the relationship between the stroke amount X and the sensor output Vs, the rotation angle θ and the signal of the stroke sensor **86** are adjusted to have a linear relationship.

As illustrated in FIGS. **3** and **6**, the housing **88** is mounted to a dash panel **9**, which is a partition wall that separates a cabin **8** from a non-cabin space **7** such as an engine compartment and the like of the vehicle **6**. In some cases, the dash panel **9** is referred to as a bulkhead. In the non-cabin space **7**, in addition to an engine of the vehicle **6**, a battery, an air conditioning device, and the like of the vehicle **6** are disposed.

As illustrated in FIG. **3**, the housing **88** is formed in a bottomed tubular shape, and includes a first mounting portion **881**, a second mounting portion **882**, a housing bottom portion **883**, and a housing tubular portion **884**. Herein, the upside with respect to the front of the vehicle **6** is simply referred to as the upside, for convenience of explanation. The downside with respect to the front of the vehicle **6** is simply referred to as the downside.

The first mounting portion **881** is connected to the housing bottom portion **883** described later, and extends upward from the housing bottom portion **883**. The first mounting portion **881** includes a first mounting hole **885**. The first mounting portion **881** is mounted to the dash panel **9** by inserting a bolt **887** into the first mounting hole **885** and a first hole **901** of the dash panel **9**. Herein, the bolt **887** is inserted so as not to penetrate through the dash panel **9**.

The second mounting portion **882** is connected to the housing tubular portion **884** described later, and extends downward from the housing tubular portion **884**. The second mounting portion **882** includes a second mounting hole **886**. The second mounting portion **882** is mounted to the dash panel **9** by inserting another bolt **887** into the second mounting hole **886** and a second hole **902** of the dash panel **9**.

The housing bottom portion **883** supports a part of the lever portion **812** such that the lever portion **812** is rotatable about the rotation axis O. The housing bottom portion **883** also supports the stroke sensor **86**.

The housing tubular portion **884** has a tubular shape, is connected to the housing bottom portion **883**, and extends downward from the housing bottom portion **883**. The housing tubular portion **884** houses a part of the lever portion **812**.

The reaction force generation portion **90** is connected to the housing tubular portion **884** and the lever portion **812**, and generates a reaction force Fr applied on the lever portion **812** in accordance with the stroke amount X. Specifically, the reaction force generation portion **90** includes an elastic member **91**.

The elastic member **91** is, for example, a helical compression spring. The elastic member **91** is connected to an inner front side of the housing tubular portion **884** and the lever portion **812**. Thus, when the brake pedal **81** is operated by the pedal force of the driver of the vehicle **6**, a force that corresponds to the pedal force is transmitted from the lever portion **812** to the elastic member **91**. As a result, the elastic member **91** is elastically deformed, that is, contracts herein, and thus a restoration force is generated. This restoration force generates the reaction force Fr applied on the lever portion **812**. The restoration force of the elastic member **91** is proportional to a deformation amount of the elastic member **91**. The deformation amount of the elastic member **91** is proportional to the stroke amount X. Thus, the restoration force of the elastic member **91** is proportional to the stroke amount X. Therefore, as illustrated in FIG. **7**, the stroke amount X and the reaction force Fr have a linear relationship. Herein, the rotation angle θ described above is also adjusted to have a linear relationship with the reaction force Fr.

The vehicle brake system **1** is configured as described above. The left front wheel FL, the right front wheel FR, the left rear wheel RL, and the right rear wheel RR of the vehicle **6** are controlled by processing of the control calculation section **633** of the ECU **53** in the vehicle brake system **1**.

Next, the processing of the control calculation section **633** will be described with reference to a flowchart of FIG. **8**. Here, for example, when an ignition of the vehicle **6** is turned on, the control calculation section **633** executes the programs stored in the ROM of the storage section **632**. In an initial state, the first power supply **401** is set as the power supply source that supplies power to the ECU **53**.

In step S**100**, the control calculation section **633** obtains various information. Specifically, the control calculation section **633** obtains a first voltage Vb**1**, which is the voltage applied from the first power supply **401** to the ECU **53**, from the first voltage sensor **451** through the communication section **631**. The control calculation section **633** also obtains the hydraulic pressure of the brake fluid flowing from the first actuator **10** to the second actuator **20**, from the first pressure sensor **14** through the communication section **631**. The control calculation section **633** further obtains the brake hydraulic pressure on the downstream side of the first differential pressure control valve **212** from the second pressure sensor **213** through the communication section **631**. The control calculation section **633** also obtains the brake hydraulic pressure on the downstream side of the second differential pressure control valve **262** from the third pressure sensor **263** through the communication section **631**. The control calculation section **633** further obtains the sensor output Vs that corresponds to the stroke amount X of the brake pedal **81** from the stroke sensor **86** through the communication section **631**. The control calculation section **633** also obtains a yaw rate, which is a rate of change in rotation angle of the vehicle **6** in the turning direction, from a yaw rate sensor (not illustrated) through the communication section **631**. The control calculation section **633** further obtains acceleration of the vehicle **6** from an acceleration sensor (not illustrated) through the communication section **631**. The control calculation section **633** also obtains a steering angle of the vehicle **6** from a steering angle sensor (not illustrated) through the communication section **631**. The control calculation section **633** further obtains wheel speeds of the left front wheel FL, the right front wheel FR, the left rear wheel RL, and the right rear wheel RR from wheel speed sensors (not illustrated) through the communication section **631**. The control calculation section **633** also obtains a vehicle speed of the vehicle **6** from a vehicle speed sensor (not illustrated) through the communication section **631**. Herein, the vehicle speed is abbreviation of the estimated vehicle body speed.

In step S**110**, the control calculation section **633** determines whether the first power supply **401** and the second power supply **402** are normal. Here, this determination will be described with reference to a flowchart of FIG. **9**.

In step S**200**, the control calculation section **633** determines whether the first power supply **401** is normal. Specifically, the control calculation section **633** determines whether the first voltage Vb**1** obtained in step S**100** is equal to or higher than a first voltage threshold Vb_th**1** and equal to or lower than a second voltage threshold Vb_th**2**. When the first voltage Vb**1** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**, the first power supply **401** is normal. Thus, the processing proceeds to step S**210**. When the first voltage Vb**1** is lower than the first voltage threshold Vb_th**1**, the first power supply **401** is abnormal. Thus, the processing proceeds to step S**240**. When the first voltage Vb**1** is higher than the second voltage threshold Vb_th**2**, the first power supply **401** is abnormal. Thus, the processing proceeds to step S**240**. The first voltage threshold Vb_th**1** and the second voltage threshold Vb_th**2** are set through experiments, simulations, or the like.

In step S**210** subsequent to step S**200**, the control calculation section **633** outputs, to the power supply switching circuit **403**, the signal for setting the second power supply **402** as the power supply source that supplies power to the ECU **53**. With this output, the power supply switching circuit **403** switches the power supply source that supplies power to the ECU **53** from the first power supply **401** to the second power supply **402**.

Subsequently, in step S**220**, the control calculation section **633** obtains a second voltage Vb**2**, which is the voltage applied from the second power supply **402** to the ECU **53**, from the second voltage sensor **452** through the communication section **631**. The control calculation section **633** determines whether the obtained second voltage Vb**2** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**. When the second voltage Vb**2** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**, the second power supply **402** is normal. Thus, the processing proceeds to step S**230**. When the second voltage Vb**2** is lower than the first voltage threshold Vb_th**1**, the second power supply **402** is abnormal. Thus, the processing proceeds to step S**240**. When the second voltage Vb**2** is higher than the second voltage threshold Vb_th**2**, the second power supply **402** is abnormal. Thus, the processing proceeds to step S**240**.

In step S**230** subsequent to step S**220**, since both of the first power supply **401** and the second power supply **402** are normal, the control calculation section **633** sets a power supply normality flag to ON, for example. Then, the processing proceeds to step S**120**.

In step S**240**, the first power supply **401** or the second power supply **402** is abnormal, and thus the control calculation section **633** determines which of the first power supply **401** or the second power supply **402** is normal. The control calculation section **633** outputs, to the power supply switching circuit **403**, the signal for switching the power supply source that supplies power to the ECU **53** to the normal one. With this output, the power supply switching circuit **403** switches the power supply source that supplies power to the ECU **53** to the one of the first power supply **401** or the second power supply **402**, which is normal. Then, the processing proceeds to step S**250**. When both of the first power supply **401** and the second power supply **402** are abnormal, the processing proceeds to step S**250**.

In step S**250** subsequent to step S**240**, since the first power supply **401** or the second power supply **402** is abnormal, the control calculation section **633** sets a power supply abnormality flag to ON, for example. Then, the processing proceeds to step S**190**.

In this manner, the control calculation section **633** determines whether the first power supply **401** and the second power supply **402** are normal. Then, when the first power supply **401** and the second power supply **402** are normal, the processing proceeds to step S**120**. When the first power supply **401** or the second power supply **402** is abnormal, the processing proceeds to step S**190**.

In step S120 subsequent to step S110, as illustrated in FIG. 8, the control calculation section 633 determines whether the control calculation section 633 itself is normal. For example, the control calculation section 633 periodically outputs a watchdog signal to a monitoring integrated circuit (IC) (not illustrated). The monitoring IC determines whether the monitoring IC has detected the watchdog signal from the control calculation section 633. Then, when the monitoring IC has detected the watchdog signal from the control calculation section 633, the monitoring IC determines that the control calculation section 633 is normal while the monitoring IC outputs a low-level signal to the control calculation section 633. When the control calculation section 633 receives the low-level signal from the monitoring IC, the control calculation section 633 is normal. Thus, the processing proceeds to step S130. When the monitoring IC does not detect the watchdog signal from the control calculation section 633, the monitoring IC outputs a signal, which is, for example, a high-level signal, indicating that the control calculation section 633 is abnormal to the control calculation section 633. When the control calculation section 633 receives the high-level signal from the monitoring IC, the control calculation section 633 is abnormal. Thus, the processing proceeds to step S185.

In step S130 subsequent to step S120, the control calculation section 633 drives the first actuator 10. Specifically, the control calculation section 633 outputs the signal for driving the first actuator 10 to the first drive circuit 71. The first drive circuit 71 drives the first actuator motor 13 based on the signal from the control calculation section 633. The first actuator motor 13 rotates based on the signal from the first drive circuit 71 to drive the first pump 12. With this driving, the first pump 12 increases the pressure of the brake fluid from the reservoir 11. The brake fluid with the increased hydraulic pressure flows from the first actuator 10 to the second actuator 20.

Subsequently, in step S140, the control calculation section 633 determines whether the first drive circuit 71 is normal. Specifically, the control calculation section 633 obtains a first hydraulic pressure P1 from the first pressure sensor 14 through the communication section 631. The first hydraulic pressure P1 is the hydraulic pressure of the brake fluid having flowed from the first actuator 10 to the second actuator 20 in step S130. Then, the control calculation section 633 determines whether the first hydraulic pressure P1 is equal to or higher than a first hydraulic pressure threshold P1_th. When the first hydraulic pressure P1 is equal to or higher than the first hydraulic pressure threshold P1_th, the first actuator 10 is driven normally by the first drive circuit 71. Thus, the first drive circuit 71 is normal. Therefore, at this time, the processing proceeds to step S150. When the first hydraulic pressure P1 is lower than the first hydraulic pressure threshold P1_th, the first actuator 10 is not driven normally by the first drive circuit 71. Thus, the first drive circuit 71 is abnormal. Therefore, at this time, the processing proceeds to step S185. The first hydraulic pressure threshold P1_th is set through experiments, simulations, or the like. Alternatively, when the first hydraulic pressure P1 is lower than the first hydraulic pressure threshold P1_th, the first actuator 10 may be determined to be abnormal.

In step S150 subsequent to step S140, the control calculation section 633 drives the second actuator 20. Specifically, the control calculation section 633 outputs the signal for driving the second actuator 20 to the second drive circuit 72. The second drive circuit 72 drives the second actuator motor 30 based on the signal from the control calculation section 633. The second actuator motor 30 rotates based on the signal from the second drive circuit 72 to drive the second pump 224 and the third pump 274.

At this time, the second pump 224 sucks the brake fluid stored in the first pressure adjustment reservoir 221. The sucked brake fluid flows between the first differential pressure control valve 212 and each of the first pressure increase control valve 215 and the second pressure increase control valve 218 after the brake fluid flows through the first return-flow pipe line 223. As a result, a pressure of the brake fluid flowing between the first differential pressure control valve 212 and each of the first pressure increase control valve 215 and the second pressure increase control valve 218 is increased.

At this time, the third pump 274 sucks a brake fluid stored in the second pressure adjustment reservoir 271. The sucked brake fluid flows between the second differential pressure control valve 262 and each of the third pressure increase control valve 265 and the fourth pressure increase control valve 268 after the brake fluid flows through the second return-flow pipe line 273. As a result, a pressure of the brake fluid flowing between the second differential pressure control valve 262 and each of the third pressure increase control valve 265 and the fourth pressure increase control valve 268 is increased.

Subsequently, in step S160, the control calculation section 633 determines whether the second drive circuit 72 is normal. Specifically, the control calculation section 633 obtains a second hydraulic pressure P2 from the second pressure sensor 213 through the communication section 631. The second hydraulic pressure P2 is the pressure of the brake fluid flowing between the first differential pressure control valve 212 and each of the first pressure increase control valve 215 and the second pressure increase control valve 218 in step S150. The control calculation section 633 also obtains a third hydraulic pressure P3 from the third pressure sensor 263 through the communication section 631. The third hydraulic pressure P3 is the pressure of the brake fluid flowing between the second differential pressure control valve 262 and each of the third pressure increase control valve 265 and the fourth pressure increase control valve 268 in step S150.

Then, the control calculation section 633 determines whether the second hydraulic pressure P2 is equal to or higher than a second hydraulic pressure threshold P2_th and the third hydraulic pressure P3 is equal to or higher than a third hydraulic pressure threshold P3_th. When the second hydraulic pressure P2 is equal to or higher than the second hydraulic pressure threshold P2_th and the third hydraulic pressure P3 is equal to or higher than the third hydraulic pressure threshold P3_th, the second actuator 20 is driven normally by the second drive circuit 72. Thus, the second drive circuit 72 is normal. Therefore, at this time, the processing proceeds to step S170. When the second hydraulic pressure P2 is lower than the second hydraulic pressure threshold P2_th or the third hydraulic pressure P3 is lower than the third hydraulic pressure threshold P3_th, the second actuator 20 is not driven normally by the second drive circuit 72. Thus, the second drive circuit 72 is abnormal. Therefore, at this time, the processing proceeds to step S185. The second hydraulic pressure threshold P2_th and the third hydraulic pressure threshold P3_th are set through experiments, simulations, or the like. Alternatively, when the second hydraulic pressure P2 is lower than the second hydraulic pressure threshold P2_th or the third hydraulic pressure P3 is lower than the third hydraulic pressure threshold P3_th, the second actuator 20 may be determined to be abnormal.

In step S170 subsequent to step S160, the control calculation section 633 determines whether the stroke sensor 86 is normal. Specifically, the control calculation section 633 determines whether the sensor output Vs obtained in step S100 is equal to or higher than a first sensor threshold Vs_th1 and equal to or lower than a second sensor threshold Vs_th2. When the sensor output Vs is equal to or higher than the first sensor threshold Vs_th1 and equal to or lower than the second sensor threshold Vs_th2, the stroke sensor 86 is normal. Thus, the processing proceeds to step S180. When the sensor output Vs is lower than the first sensor threshold Vs_th1, the stroke sensor 86 is abnormal. Thus, the processing proceeds to step S185. When the sensor output Vs is higher than the second sensor threshold Vs_th2, the stroke sensor 86 is abnormal. Thus, the processing proceeds to step S185. The first sensor threshold Vs_th1 is set based on, for example, an initial position of the brake pedal 81 and a variation in position of the brake pedal 81. The second sensor threshold Vs_th2 is set based on, for example, the maximum value of the stroke amount X of the brake pedal 81 and a variation in position of the brake pedal 81.

In step S180, the control calculation section 633 performs normal control of the first actuator 10 based on the sensor output Vs that corresponds to the stroke amount X, obtained in step S100.

For example, as illustrated in FIG. 10, when the pedal portion 811 is pressed by the driver of the vehicle 6, the lever portion 812 rotates about the rotation axis O. With this rotation, the stroke amount X increases, and thus the sensor output Vs increases. At this time, the control calculation section 633 outputs, to the first drive circuit 71, the signal for driving the first actuator 10 to increase the first hydraulic pressure P1 in order to decelerate the vehicle 6. The first drive circuit 71 drives the first actuator motor 13 based on the signal from the control calculation section 633. At this time, a rotation speed of the first actuator motor 13 increases. With this increase in the rotation speed, the first pump 12 increases the pressure of the brake fluid from the reservoir 11. Therefore, the first hydraulic pressure P1 increases. The brake fluid that is relatively high in the first hydraulic pressure P1 flows from the first actuator 10 to the second actuator 20.

When the stroke amount X increases, the elastic member 91 contracts because the elastic member 91 is connected to the inner front side of the housing tubular portion 884 and the lever portion 812. With this contraction, the reaction force Fr is generated along with the restoration force of the elastic member 91. When the driver of the vehicle 6 takes his or her foot off the pedal portion 811, the reaction force Fr causes the brake pedal 81 to return to the initial position. In FIG. 10, the position of the brake pedal 81 in the initial state is indicated by a dashed double-dotted line. Herein, the stroke amount X is the amount of the translational movement of the pedal portion 811 toward the front of the vehicle 6, and thus a direction of the reaction force Fr is a direction toward the rear.

In step S180, the control calculation section 633 performs the normal control, ABS control, VSC control, and the like. The ABS is an abbreviation for Antilock Brake System. The VSC is an abbreviation for Vehicle Stability Control.

For example, when the pedal portion 811 is pressed by the driver of the vehicle 6, the control calculation section 633 performs the normal control, which is brake control made by the driver of the vehicle 6 operating the brake pedal 81. At this time, the lever portion 812 rotates about the rotation axis O. With this rotation, the stroke amount X increases, and thus the sensor output Vs increases. At this time, the control calculation section 633 controls the second drive circuit 72 in order to decelerate the vehicle 6. With this control, the second drive circuit 72 brings the pressure increase control valves in the second actuator 20 into the communicating state by bringing the solenoid coils of the pressure increase control valves in the second actuator 20 into the non-energized state. Therefore, the brake fluid having flowed from the first actuator 10 to the second actuator 20 flows to each of the left front wheel W/C 2, the right front wheel W/C 3, the left rear wheel W/C 4, and the right rear wheel W/C 5, through each corresponding pressure increase control valve. Thus, each brake pad (not illustrated) comes into frictional contact with the corresponding brake disc. Therefore, each wheel that corresponds to each brake disc is decelerated, and thus the vehicle 6 decelerates. As a result, the vehicle 6 stops.

The control calculation section 633 calculates a slip ratio of each of the left front wheel FL, the right front wheel FR, the left rear wheel RL, and the right rear wheel RR, based on, for example, each of the wheel speeds and the vehicle speed obtained in step S100. Then, the control calculation section 633 determines whether to perform the ABS control based on the slip ratios. When the ABS control is performed, the control calculation section 633 performs any of a pressure reducing mode, a maintaining mode, and a pressure increasing mode in accordance with the slip ratios. In the pressure reducing mode, the pressure increase control valve that corresponds to the control target wheel is brought into the blocking state while the pressure reduction control valve is appropriately brought into the communicating state. Thus, the pressure is reduced in the W/C that corresponds to the control target wheel. In the maintaining mode, the pressure increase control valve and the pressure reduction control valve that correspond to the control target wheel are brought into the blocking states. Thus, the pressure is maintained in the W/C that corresponds to the control target wheel. In the pressure increasing mode, the pressure reduction control valve that corresponds to the control target wheel is brought into the blocking state while the pressure increase control valve is appropriately brought into the communicating state. Thus, the pressure is increased in the W/C that corresponds to the control target wheel. The slip ratio of each wheel of the vehicle 6 is controlled in this manner. Thus, the left front wheel FL, the right front wheel FR, the left rear wheel RL, and the right rear wheel RR are prevented from being locked.

The control calculation section 633 also calculates a skid state of the vehicle 6 based on, for example, the yaw rate, the steering angle, the acceleration, each of the wheel speeds, the vehicle speed, and the like obtained in step S100. Then, the control calculation section 633 determines whether to perform the VSC control based on the skid state of the vehicle 6. When the VSC control is performed, the control calculation section 633 selects a control target wheel that is used for stabilizing turning of the vehicle 6, based on the skid state of the vehicle 6. The control calculation section 633 controls the second drive circuit 72 such that the pressure is increased in the W/C that corresponds to the selected control target wheel. At this time, the second drive circuit 72 drives the pump that corresponds to the control target wheel by driving the second actuator motor 30. With this driving, the pump that corresponds to the control target wheel sucks the brake fluid stored in the pressure adjustment reservoir that corresponds to the control target wheel. This sucked brake fluid flows to the W/C that corresponds to the control target wheel after the brake fluid flows through the return-flow pipe line that corresponds to the control target wheel. As a result, the brake hydraulic pressure is increased in the W/C that corresponds to the control target wheel. Thus, the skid of the vehicle **6** is suppressed. Therefore, traveling of the vehicle **6** is stabilized.

In this manner, in step S**180**, the control calculation section **633** performs the normal control, the ABS control, the VSC control, and the like. Then, the processing returns to step S**100**. In step S**180**, in addition to the normal control, the ABS control, and the VSC control as described above, the control calculation section **633** may perform collision avoidance control, regeneration cooperative control, and the like based on signals from another ECU (not illustrated).

In step S**185**, when the control calculation section **633** itself is abnormal, the control calculation section **633** outputs a signal indicating that the control calculation section **633** is abnormal to a notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the control calculation section **633** is abnormal through screen display, sound, light, and the like.

When the first drive circuit **71** is abnormal, the control calculation section **633** outputs a signal indicating that the first drive circuit **71** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first drive circuit **71** is abnormal through screen display, sound, light, and the like.

When the second drive circuit **72** is abnormal, the control calculation section **633** outputs a signal indicating that the second drive circuit **72** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second drive circuit **72** is abnormal through screen display, sound, light, and the like.

When the stroke sensor **86** is abnormal, the control calculation section **633** outputs a signal indicating that the stroke sensor **86** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the stroke sensor **86** is abnormal through screen display, sound, light, and the like. After step S**185**, the processing proceeds to step S**190**.

In step S**190**, when the first power supply **401** and the second power supply **402** are abnormal, the control calculation section **633** cannot control the first drive circuit **71** and the second drive circuit **72** normally. When the control calculation section **633** itself is abnormal, the control calculation section **633** cannot control the first drive circuit **71** and the second drive circuit **72** normally. Therefore, the vehicle **6** is controlled to decelerate and stop in order to ensure safety of the vehicle **6** by another calculation section or the like that is different from the control calculation section **633**. For example, in these cases, a regenerative brake (not illustrated), a parking brake (not illustrated), and the like are controlled by another ECU that is different from the ECU **53**. As a result, the vehicle **6** safely decelerates and stops.

When the first power supply **401**, the second power supply **402**, and the control calculation section **633** are normal, even if one of the first drive circuit **71** or the second drive circuit **72** fails, the control calculation section **633** can control the normal one. Thus, the control calculation section **633** controls either the first drive circuit **71** or second drive circuit **72**, which is normal, to cause the vehicle **6** to decelerate and stop.

For example, when the first power supply **401**, the second power supply **402**, the control calculation section **633**, and the first drive circuit **71** are normal while the second drive circuit **72** is abnormal, the control calculation section **633** causes the vehicle **6** to decelerate and stop. Specifically, the control calculation section **633** outputs the signal for driving the first actuator **10** to the first drive circuit **71**. The first drive circuit **71** drives the first actuator motor **13** based on the signal from the control calculation section **633**. The first actuator motor **13** rotates based on the signal from the first drive circuit **71** to drive the first pump **12**. The first pump **12** increases the pressure of the brake fluid from the reservoir **11**. The brake fluid with the increased hydraulic pressure flows to the second actuator **20**. The brake fluid having flowed to the second actuator **20** flows to each of the left front wheel W/C **2**, the right front wheel W/C **3**, the left rear wheel W/C **4**, and the right rear wheel W/C **5**, through each corresponding pressure increase control valve. With this flow, each brake pad (not illustrated) comes into frictional contact with the corresponding brake disc. Therefore, each wheel that corresponds to each brake disc is decelerated, and thus the vehicle **6** decelerates. As a result, the vehicle **6** stops.

For example, when the first power supply **401**, the second power supply **402**, the control calculation section **633**, and the second drive circuit **72** are normal while the first drive circuit **71** is abnormal, the control calculation section **633** causes the vehicle **6** to decelerate and stop. Specifically, the control calculation section **633** outputs the signal for driving the second actuator **20** to the second drive circuit **72**. The second drive circuit **72** drives the second actuator motor **30** based on the signal from the second control calculation section **623**. The second actuator motor **30** rotates based on the signal from the second drive circuit **72** to drive the second pump **224** and the third pump **274**.

At this time, the second pump **224** sucks the brake fluid stored in the first pressure adjustment reservoir **221**. The sucked brake fluid flows between the first differential pressure control valve **212** and each of the first pressure increase control valve **215** and the second pressure increase control valve **218** after the brake fluid flows through the first return-flow pipe line **223**. The brake fluid caused to flow by the second pump **224** flows to the left front wheel W/C **2** through the first pressure increase control valve **215**. The brake fluid caused to flow by the second pump **224** flows to the right front wheel W/C **3** through the second pressure increase control valve **218**.

At this time, the third pump **274** sucks the brake fluid stored in the second pressure adjustment reservoir **271**. The sucked brake fluid flows between the second differential pressure control valve **262** and each of the third pressure increase control valve **265** and the fourth pressure increase control valve **268** after the brake fluid flows through the second return-flow pipe line **273**. The brake fluid caused to flow by the third pump **274** flows to the right rear wheel W/C **5** through the third pressure increase control valve **265**. The brake fluid caused to flow by the third pump **274** flows to the left rear wheel W/C **4** through the fourth pressure increase control valve **268**.

Thus, each brake pad (not illustrated) comes into frictional contact with the corresponding brake disc. Therefore, each wheel that corresponds to each brake disc is decelerated, and thus the vehicle **6** decelerates. As a result, the vehicle **6** stops. After step **S190**, the processing returns to step **S100**.

In this manner, the processing of the control calculation section **633** is executed.

In the vehicle brake system **1**, the brake of the vehicle **6** is controlled as described above. In the vehicle brake system **1**, redundancy is improved. Hereinafter, the improvement of the redundancy will be described. Herein, the redundancy refers to safety, which is obtained by the vehicle brake system **1** provided with backup devices in preparation for a case where any fault occurs in the vehicle brake system **1**.

In the present embodiment, the ECU **53** includes the first drive circuit **71** and the second drive circuit **72**. The first drive circuit **71** drives the first actuator **10**. The second drive circuit **72** drives the second actuator **20**. With this configuration, even if one of the first drive circuit **71** or the second drive circuit **72** fails, the vehicle brake system **1** can cause the vehicle **6** to decelerate and stop safely by using the other normal one. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

The vehicle brake system **1** also has effects as described below.

[1] The vehicle brake system **1** includes the first power supply **401** and the second power supply **402**. With this configuration, even if one of the first power supply **401** or the second power supply **402** fails, power can be ensured by using the other normal one. Thus, the vehicle brake system **1** can cause the vehicle **6** to decelerate and stop safely. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

[2] The vehicle brake system **1** includes the first actuator **10** and the second actuator **20**. With this configuration, even if one of the first actuator **10** or the second actuator **20** fails, the vehicle brake system **1** can cause the vehicle **6** to decelerate and stop safely by using the other normal one. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

Second Embodiment

In a second embodiment, the ECU **53** of the vehicle brake system **1** includes two microcomputers. The vehicle brake system **1** further includes two voltage sensors. The vehicle brake device **80** includes two stroke sensors. Processing executed by the ECU **53** is different. The others are similar to those of the first embodiment.

As illustrated in FIG. **11**, the ECU **53** of the vehicle brake system **1** according to the second embodiment includes a first microcomputer **61** and a second microcomputer **62**, in addition to the first drive circuit **71** and the second drive circuit **72** described above.

The first microcomputer **61** corresponds to a first hydraulic pressure control unit, and controls the first actuator **10** by controlling the first drive circuit **71**. Specifically, the first microcomputer **61** includes a first communication section **611**, a first storage section **612**, and a first control calculation section **613**.

The first communication section **611** includes an interface for communicating with the first pressure sensor **14**. The first communication section **611** also includes an interface for communicating with the first voltage sensor **451** and an interface for communicating with the second voltage sensor **452**. The first communication section **611** further includes an interface for communicating with the second microcomputer **62** described later. The first communication section **611** also includes an interface for communicating with a first stroke sensor **861** described later and an interface for communicating with a second stroke sensor **862** described later.

The first storage section **612** includes non-volatile memories such as a ROM and a flash memory, and a volatile memory such as a RAM. The non-volatile memories and the volatile memory are non-transitory tangible storage media.

The first control calculation section **613** includes a CPU and the like. The first control calculation section **613** outputs a signal for driving the first actuator motor **13** to the first drive circuit **71** by executing a program stored in the ROM of the first storage section **612**.

The second microcomputer **62** corresponds to a second hydraulic pressure control unit, and controls the second actuator **20** by controlling the second drive circuit **72**. Specifically, the second microcomputer **62** includes a second communication section **621**, a second storage section **622**, and a second control calculation section **623**.

The second communication section **621** includes an interface for communicating with the second pressure sensor **213** and an interface for communicating with the third pressure sensor **263**. The second communication section **621** also includes an interface for communicating with the first voltage sensor **451** and an interface for communicating with the second voltage sensor **452**. The second communication section **621** further includes an interface for communicating with the first communication section **611** of the first microcomputer **61**. The second communication section **621** also includes an interface for communicating with the first stroke sensor **861** described later and an interface for communicating with the second stroke sensor **862** described later.

The second storage section **622** includes non-volatile memories such as a ROM and a flash memory, and a volatile memory such as a RAM. The non-volatile memories and the volatile memory are non-transitory tangible storage media.

The second control calculation section **623** includes a CPU and the like. The second control calculation section **623** outputs a signal for driving the second actuator motor **30** to the second drive circuit **72** by executing a program stored in the ROM of the second storage section **622**.

The vehicle brake device **80** includes the brake pedal **81**, the housing **88**, and the reaction force generation portion **90**, which are similar to those of the first embodiment described above. The vehicle brake device **80** also includes a first sensor power supply line **821**, a first sensor ground line **831**, a first sensor output line **841**, and a second sensor output line **842**. The vehicle brake device **80** further includes a second sensor power supply line **822**, a second sensor ground line **832**, a third sensor output line **843**, a fourth sensor output line **844**, the first stroke sensor **861**, and the second stroke sensor **862**.

As illustrated in FIGS. **11** and **12**, the first sensor power supply line **821** is connected to the ECU **53** and the first stroke sensor **861**. With this connection, power from the first power supply **401** and the second power supply **402** is supplied to the first stroke sensor **861** through the ECU **53**.

The first sensor ground line **831** is connected to the ECU **53** and the first stroke sensor **861**.

The first sensor output line **841** is connected to the ECU **53** and the first stroke sensor **861**.

The second sensor output line **842** is connected to the ECU **53** and the second stroke sensor **862**.

The second sensor power supply line **822** is connected to the ECU **53** and the second stroke sensor **862**. With this connection, power from the first power supply **401** and the second power supply **402** is supplied to the second stroke sensor **862** through the ECU **53**.

The second sensor ground line **832** is connected to the ECU **53** and the second stroke sensor **862**.

The third sensor output line **843** is connected to the ECU **53** and the first stroke sensor **861**.

The fourth sensor output line **844** is connected to the ECU **53** and the second stroke sensor **862**.

The first stroke sensor **861** outputs a signal being in accordance with the stroke amount X of the brake pedal **81** to the first microcomputer **61** through the first sensor output line **841**. The first stroke sensor **861** also outputs the signal being in accordance with the stroke amount X of the brake pedal **81** to the second microcomputer **62** through the third sensor output line **843**. Herein, as illustrated in FIG. **13**, a first sensor output Vs**1**, which is the signal from the first stroke sensor **861**, is adjusted to be constant at V**1** when the stroke amount X is smaller than X**1**. The first sensor output Vs**1** is adjusted to increase as the stroke amount X increases when the stroke amount X is equal to or larger than X**1** and smaller than X**2**. The first sensor output Vs**1** is adjusted to be constant at V**2**, which is a value higher than V**1**, when the stroke amount X is equal to or larger than X**2**. Note that V**1** and V**2** are set through experiments, simulations, or the like. Herein, V**1** is set to a value lower than the first sensor threshold Vs_th**1** described above. Further, V**2** is set to a value higher than the second sensor threshold Vs_th**2** described above.

As illustrated in FIGS. **11** and **12**, the second stroke sensor **862** outputs a signal being in accordance with the stroke amount X of the brake pedal **81** to the first microcomputer **61** through the second sensor output line **842**. The second stroke sensor **862** also outputs the signal being in accordance with the stroke amount X of the brake pedal **81** to the second microcomputer **62** through the fourth sensor output line **844**. Herein, as illustrated in FIG. **13**, a second sensor output Vs**2**, which is the signal from the second stroke sensor **862**, is adjusted to be constant at V**2**, which is the value higher than V**1**, when the stroke amount X is smaller than X**1**. The second sensor output Vs**2** is adjusted to decrease as the stroke amount X increases when the stroke amount X is equal to or larger than X**1** and smaller than X**2**. The second sensor output Vs**2** is adjusted to be constant at V**1**, which is the value lower than V**2**, when the stroke amount X is equal to or larger than X**2**.

The vehicle brake system **1** according to the second embodiment is configured as described above.

Next, processing of the first control calculation section **613** will be described with reference to a flowchart of FIG. **14**. Here, for example, when the ignition of the vehicle **6** is turned on, the first control calculation section **613** executes programs stored in the ROM of the first storage section **612**. As with the above description, in the initial state, the first power supply **401** is set as the power supply source that supplies power to the ECU **53**.

In step S**300**, the first control calculation section **613** obtains various information. Specifically, the first control calculation section **613** obtains the first voltage Vb**1**, which is the voltage applied from the first power supply **401** to the ECU **53**, from the first voltage sensor **451** through the first communication section **611**. The first control calculation section **613** also obtains the hydraulic pressure of the brake fluid supplied from the first actuator **10** to the second actuator **20**, from the first pressure sensor **14** through the first communication section **611**. The first control calculation section **613** further obtains the first sensor output Vs**1** that corresponds to the stroke amount X of the brake pedal **81** from the first stroke sensor **861** through the first sensor output line **841** and the first communication section **611**. The first control calculation section **613** also obtains the second sensor output Vs**2** that corresponds to the stroke amount X of the brake pedal **81** from the second stroke sensor **862** through the second sensor output line **842** and the first communication section **611**.

Subsequently, in step S**310**, the first control calculation section **613** determines whether the first power supply **401** and the second power supply **402** are normal, similarly to step S**110** executed by the control calculation section **633**, which has been described above. When the first power supply **401** and the second power supply **402** are normal, the processing proceeds to step S**320**. When the first power supply **401** or the second power supply **402** is abnormal, the processing proceeds to step S**390**.

In step S**320** subsequent to step S**310**, the first control calculation section **613** determines whether the second control calculation section **623** is normal. Specifically, the first control calculation section **613** determines whether the first control calculation section **613** has received a signal indicating that the second control calculation section **623** is abnormal, which will be described later, from the second control calculation section **623**. When the second control calculation section **623** is normal, the processing proceeds to step S**330**. When the second control calculation section **623** is abnormal, the processing proceeds to step S**390**.

In step S**330** subsequent to step S**320**, the first control calculation section **613** determines whether the first control calculation section **613** itself is normal by using the watchdog signal and the monitoring IC, similarly to step S**120** executed by the control calculation section **633**, which has been described above. When the first control calculation section **613** itself is normal, the processing proceeds to step S**340**. When the first control calculation section **613** itself is abnormal, the processing proceeds to step S**385**.

In step S**340** subsequent to step S**330**, the first control calculation section **613** drives the first actuator **10**, similarly to step S**130** executed by the control calculation section **633**, which has been described above.

Subsequently, in step S**350**, the first control calculation section **613** determines whether the first drive circuit **71** is normal based on the first hydraulic pressure P**1**, similarly to step S**140** executed by the control calculation section **633**, which has been described above. When the first drive circuit **71** is normal, the processing proceeds to step S**355**. When the first drive circuit **71** is abnormal, the processing proceeds to step S**385**. As described above, the first hydraulic pressure P**1** is the hydraulic pressure of the brake fluid having flowed from the first actuator **10** to the second actuator **20**.

In step S**355** subsequent to step S**350**, the first control calculation section **613** determines whether the first sensor power supply line **821** and the second sensor power supply line **822** are normal. Specifically, the first control calculation section **613** determines whether each of the first sensor output Vs**1** and the second sensor output Vs**2** obtained in step S**300** is equal to or higher than V**1** and equal to or lower than V**2**. As described above, V**1** and V**2** are set through experiments, simulations, or the like. Herein, as will be described later, V**1** is set to a value lower than the first sensor threshold Vs_th**1** described above. Further, V**2** is set to a value higher than the second sensor threshold Vs_th**2** described above.

When the first sensor output Vs**1** is equal to or higher than V**1** and equal to or lower than V**2**, the first sensor power supply line **821** is normal. When the second sensor output Vs**2** is equal to or higher than V**1** and equal to or lower than

V2, the second sensor power supply line **822** is normal. Therefore, when each of the first sensor output Vs**1** and the second sensor output Vs**2** is equal to or higher than V**1** and equal to or lower than V**2**, the first sensor power supply line **821** and the second sensor power supply line **822** are normal. Thus, the processing proceeds to step S**360**. When the first sensor output Vs**1** or the second sensor output Vs**2** is lower than V**1**, the first sensor power supply line **821** or the second sensor power supply line **822** is abnormal. In this case, for example, disconnection is made. Thus, the processing proceeds to step S**385**. When the first sensor output Vs**1** or the second sensor output Vs**2** is higher than V**2**, the first sensor power supply line **821** or the second sensor power supply line **822** is abnormal. In this case, for example, disconnection is made. Thus, the processing proceeds to step S**385**.

In step S**360** subsequent to step S**355**, the first control calculation section **613** determines whether the first stroke sensor **861** is normal. Specifically, the first control calculation section **613** determines whether the first sensor output Vs**1** obtained in step S**300** is equal to or higher than the first sensor threshold Vs_th**1** and equal to or lower than the second sensor threshold Vs_th**2**. As described above, the first sensor threshold Vs_th**1** is set based on, for example, the initial position of the brake pedal **81** and the variation in position of the brake pedal **81**. The second sensor threshold Vs_th**2** is set based on, for example, the maximum value of the stroke amount X of the brake pedal **81** and the variation in position of the brake pedal **81**. Herein, when the stroke amount X is a value near X**1** and is larger than X**1**, that is, when X>X**1**, the brake pedal **81** is placed at the initial position. When the stroke amount X is a value near X**2** and smaller than X**2**, that is, when X**2**>X, the stroke amount X of the brake pedal **81** becomes the maximum value. Therefore, herein, V**1**, V**2**, the first sensor threshold Vs_th**1**, and the second sensor threshold Vs_th**2** have a relationship of V**1**<Vs_th**1**<Vs_th**2**<V**2**.

When the first sensor output Vs**1** is equal to or higher than the first sensor threshold Vs_th**1** and equal to or lower than the second sensor threshold Vs_th**2**, the first stroke sensor **861** is normal. Thus, the processing proceeds to step S**365**. When the first sensor output Vs**1** is equal to or higher than V**1** and lower than the first sensor threshold Vs_th**1**, the first stroke sensor **861** is abnormal. Thus, the processing proceeds to step S**385**. When the first sensor output Vs**1** is higher than the second sensor threshold Vs_th**2** and equal to or lower than V**2**, the first stroke sensor **861** is abnormal. Thus, the processing proceeds to step S**385**. At this time, similarly to the first stroke sensor **861**, the first control calculation section **613** may determine whether the second stroke sensor **862** is normal based on the second sensor output Vs**2** obtained in step S**300**.

In step S**365** subsequent to step S**360**, the first control calculation section **613** determines whether the first stroke sensor **861** and the second stroke sensor **862** are normal. Specifically, the first control calculation section **613** calculates a sensor output difference Vs**1**-Vs**2**, which is a relative value of a difference between the first sensor output Vs**1** and the second sensor output Vs**2**, based on the first sensor output Vs**1** and the second sensor output Vs**2** obtained in step S**300**.

As described above, the first sensor output Vs**1** is adjusted to be constant at V**1** when the stroke amount X is smaller than X**1**, as illustrated in FIG. **13**. The first sensor output Vs**1** is adjusted to increase as the stroke amount X increases when the stroke amount X is equal to or larger than X**1** and smaller than X**2**. The first sensor output Vs**1** is constant at V**2** when the stroke amount X is equal to or larger than X**2**. The second sensor output Vs**2** is adjusted to be constant at V**2**, which is the value higher than V**1**, when the stroke amount X is smaller than X**1**. The second sensor output Vs**2** is adjusted to decrease as the stroke amount X increases when the stroke amount X is equal to or larger than X**1** and smaller than X**2**. The second sensor output Vs**2** is adjusted to be constant at V**1**, which is the value lower than V**2**, when the stroke amount X is equal to or larger than X**2**. Thus, when the first sensor output Vs**1** and the second sensor output Vs**2** are normal, the sensor output difference Vs**1**-Vs**2** is a value defined by the stroke amount X.

Therefore, the first control calculation section **613** determines whether the calculated sensor output difference Vs**1**-Vs**2** is within the range of the defined value described above. When the sensor output difference Vs**1**-Vs**2** is within the range of the defined value described above, the first sensor output Vs**1** and the second sensor output Vs**2** are normal. Thus, the processing proceeds to step S**370**. When the sensor output difference Vs**1**-Vs**2** is outside the range of the defined value described above, the first sensor output Vs**1** or the second sensor output Vs**2** is abnormal. At this time, the first control calculation section **613** determines which of the first stroke sensor **861** or the second stroke sensor **862** is abnormal. Then, the processing proceeds to step S**385**.

In step S**370** subsequent to step S**365**, the first control calculation section **613** determines whether the second drive circuit **72** is normal. Specifically, the first control calculation section **613** determines whether the first control calculation section **613** has received a signal indicating that the second drive circuit **72** is abnormal, which will be described later, from the second control calculation section **623**. When the second drive circuit **72** is normal, the processing proceeds to step S**380**. When the second drive circuit **72** is abnormal, the processing proceeds to step S**390**.

In step S**380** subsequent to step S**370**, the first control calculation section **613** controls the first actuator **10**, similarly to step S**180** executed by the control calculation section **633**, which has been described above. Specifically, the first control calculation section **613** performs the normal control of the first actuator **10** based on the first sensor output Vs**1** that corresponds to the stroke amount X and that has been obtained in step S**300**. At this time, the first control calculation section **613** may perform the normal control of the first actuator **10** based on the second sensor output Vs**2** that corresponds to the stroke amount X and that has been obtained in step S**300**. Then, the processing returns to step S**300**.

In step S**385**, when the first control calculation section **613** itself is abnormal, the first control calculation section **613** outputs a signal indicating that the first control calculation section **613** is abnormal to the second control calculation section **623**. At this time, the first control calculation section **613** also outputs the signal indicating that the first control calculation section **613** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first control calculation section **613** is abnormal through screen display, sound, light, and the like.

When the first drive circuit **71** is abnormal, the first control calculation section **613** outputs a signal indicating that the first drive circuit **71** is abnormal to the second control calculation section **623**. At this time, the first control calculation section **613** outputs the signal indicating that the first drive circuit **71** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first drive circuit **71** is abnormal through screen display, sound, light, and the like.

When the first sensor power supply line **821** is abnormal, the first control calculation section **613** outputs a signal indicating that the first sensor power supply line **821** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first sensor power supply line **821** is abnormal through screen display, sound, light, and the like.

When the second sensor power supply line **822** is abnormal, the first control calculation section **613** outputs a signal indicating that the second sensor power supply line **822** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second sensor power supply line **822** is abnormal through screen display, sound, light, and the like.

When the first stroke sensor **861** is abnormal, the first control calculation section **613** outputs a signal indicating that the first stroke sensor **861** is abnormal to the notification device. When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first stroke sensor **861** is abnormal through screen display, sound, light, and the like.

When the second stroke sensor **862** is abnormal, the first control calculation section **613** outputs a signal indicating that the second stroke sensor **862** is abnormal to the notification device. When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second stroke sensor **862** is abnormal through screen display, sound, light, and the like. After step S**385**, the processing proceeds to step S**390**.

Subsequently, in step S**390**, when the first power supply **401** and the second power supply **402** are abnormal, when the first control calculation section **613** itself is abnormal, or when the first drive circuit **71** is abnormal, the first control calculation section **613** cannot control the first drive circuit **71** normally. Therefore, the vehicle **6** is controlled to decelerate and stop in order to ensure the safety of the vehicle **6** by another calculation section or the like that is different from the first control calculation section **613**.

For example, in these cases, when the second control calculation section **623** and the second drive circuit **72** are normal, the second control calculation section **623** controls the second actuator **20** by controlling the second drive circuit **72**, as described later. As a result, the vehicle **6** safely decelerates and stops.

In these cases, when the second control calculation section **623** and the second drive circuit **72** are abnormal, the regenerative brake (not illustrated), the parking brake (not illustrated), and the like are controlled by another ECU that is different from the ECU **53**. As a result, the vehicle **6** safely decelerates and stops.

For example, when the first power supply **401**, the second power supply **402**, the first control calculation section **613**, and the first drive circuit **71** are normal while the second control calculation section **623** or the second drive circuit **72** is abnormal, the first control calculation section **613** causes the vehicle **6** to decelerate and stop.

Specifically, the first control calculation section **613** outputs the signal for driving the first actuator **10** to the first drive circuit **71**. The first drive circuit **71** drives the first actuator motor **13** based on the signal from the first control calculation section **613**. The first actuator motor **13** rotates based on the signal from the first drive circuit **71** to drive the first pump **12**.

At this time, the first pump **12** increases the pressure of the brake fluid from the reservoir **11**. The brake fluid with the increased hydraulic pressure flows to the second actuator **20**. The brake fluid having flowed to the second actuator **20** flows to each of the left front wheel W/C **2**, the right front wheel W/C **3**, the left rear wheel W/C **4**, and the right rear wheel W/C **5**, through each corresponding pressure increase control valve. With this flow, each brake pad (not illustrated) comes into frictional contact with the corresponding brake disc. Therefore, each wheel that corresponds to each brake disc is decelerated, and thus the vehicle **6** decelerates. As a result, the vehicle **6** stops. After step S**390**, the processing returns to step S**300**.

In this manner, the processing of the first control calculation section **613** is executed.

Next, processing of the second control calculation section **623** will be described with reference to a flowchart of FIG. **15**. Here, for example, when the ignition of the vehicle **6** is turned on, the second control calculation section **623** executes programs stored in the ROM of the second storage section **622**. As with the above description, in the initial state, the first power supply **401** is set as the power supply source that supplies power to the ECU **53**.

In step S**400**, the second control calculation section **623** obtains various information. Specifically, the second control calculation section **623** obtains the first voltage Vb**1**, which is the voltage applied from the first power supply **401** to the ECU **53**, from the first voltage sensor **451** through the second communication section **621**. The second control calculation section **623** also obtains the brake hydraulic pressure on the downstream side of the first differential pressure control valve **212** from the second pressure sensor **213** through the second communication section **621**. The second control calculation section **623** further obtains the brake hydraulic pressure on the downstream side of the second differential pressure control valve **262** from the third pressure sensor **263** through the second communication section **621**. The second control calculation section **623** also obtains the first sensor output Vs**1** that corresponds to the stroke amount X of the brake pedal **81** from the first stroke sensor **861** through the third sensor output line **843** and the second communication section **621**. The second control calculation section **623** further obtains the second sensor output Vs**2** that corresponds to the stroke amount X of the brake pedal **81** from the second stroke sensor **862** through the fourth sensor output line **844** and the second communication section **621**. Similarly to step S**100** executed by the control calculation section **633**, the second control calculation section **623** obtains the yaw rate, the acceleration, the steering angle, each of the wheel speeds, and the vehicle speed from each sensor (not illustrated) through the second communication section **621**.

Subsequently, in step S**410**, the second control calculation section **623** determines whether the first power supply **401** and the second power supply **402** are normal, similarly to step S**110** executed by the control calculation section **633** and step S**310** executed by the first control calculation section **613**, which have been described above. When the first power supply **401** and the second power supply **402** are normal, the processing proceeds to step S**420**. When the first power supply **401** or the second power supply **402** is abnormal, the processing proceeds to step S**490**.

In step S**420** subsequent to step S**410**, the second control calculation section **623** determines whether the first control calculation section **613** is normal. Specifically, the second control calculation section **623** determines whether the second control calculation section **623** has received, from the first control calculation section **613**, the signal indicating that the first control calculation section **613** is abnormal, output by the first control calculation section **613** in step S**385**, which has been described above. When the first control calculation section **613** is normal, the processing proceeds to step S**430**. When the first control calculation section **613** is abnormal, the processing proceeds to step S**490**.

In step S**430** subsequent to step S**420**, the second control calculation section **623** determines whether the second control calculation section **623** itself is normal, similarly to step S**120** executed by the control calculation section **633** and step S**330** executed by the first control calculation section **613**, which have been described above. Specifically, the second control calculation section **623** determines whether the second control calculation section **623** itself is normal by using the watchdog signal and the monitoring IC. When the second control calculation section **623** itself is normal, the processing proceeds to step S**440**. When the second control calculation section **623** itself is abnormal, the processing proceeds to step S**485**.

In step S**440** subsequent to step S**430**, the second control calculation section **623** drives the second actuator **20**, similarly to step S**150** executed by the control calculation section **633**, which has been described above.

Subsequently, in step S**450**, the second control calculation section **623** determines whether the second drive circuit **72** is normal based on the second hydraulic pressure P**2** and the third hydraulic pressure P**3**, similarly to step S**160** executed by the control calculation section **633**, which has been described above. When the second drive circuit **72** is normal, the processing proceeds to step S**455**. When the second drive circuit **72** is abnormal, the processing proceeds to step S**485**. As described above, the second hydraulic pressure P**2** is the pressure of the brake fluid flowing between the first differential pressure control valve **212** and each of the first pressure increase control valve **215** and the second pressure increase control valve **218**. As described above, the third hydraulic pressure P**3** is the pressure of the brake fluid flowing between the second differential pressure control valve **262** and each of the third pressure increase control valve **265** and the fourth pressure increase control valve **268**.

In step S**455** subsequent to step S**450**, the second control calculation section **623** determines whether the first sensor power supply line **821** and the second sensor power supply line **822** are normal, similarly to step S**355** executed by the first control calculation section **613**, which has been described above. Specifically, the second control calculation section **623** determines whether the first sensor power supply line **821** and the second sensor power supply line **822** are normal based on the first sensor output Vs**1** and the second sensor output Vs**2** obtained in step S**400**. When the first sensor power supply line **821** and the second sensor power supply line **822** are normal, the processing proceeds to step S**460**. When the first sensor power supply line **821** or the second sensor power supply line **822** is abnormal, the processing proceeds to step S**485**.

In step S**460** subsequent to step S**455**, the second control calculation section **623** determines whether the first stroke sensor **861** is normal, similarly to step S**360** executed by the first control calculation section **613**, which has been described above. Specifically, the second control calculation section **623** determines whether the first sensor output Vs**1** obtained in step S**400** is equal to or higher than the first sensor threshold Vs_th**1** and equal to or lower than the second sensor threshold Vs_th**2**. When the first sensor output Vs**1** is equal to or higher than the first sensor threshold Vs_th**1** and equal to or lower than the second sensor threshold Vs_th**2**, the first stroke sensor **861** is normal. Thus, the processing proceeds to step S**465**. When the first sensor output Vs**1** is lower than the first sensor threshold Vs_th**1**, the first stroke sensor **861** is abnormal. Thus, the processing proceeds to step S**485**. When the first sensor output Vs**1** is higher than the second sensor threshold Vs_th**2**, the first stroke sensor **861** is abnormal. Thus, the processing proceeds to step S**485**. At this time, similarly to the first stroke sensor **861**, the second control calculation section **623** may determine whether the second stroke sensor **862** is normal based on the second sensor output Vs**2** obtained in step S**400**.

In step S**465** subsequent to step S**460**, the second control calculation section **623** determines whether the first stroke sensor **861** and the second stroke sensor **862** are normal, similarly to step S**365** executed by the first control calculation section **613**, which has been described above. Specifically, the second control calculation section **623** determines whether the first stroke sensor **861** and the second stroke sensor **862** are normal based on the sensor output difference Vs**1**-Vs**2**. When the first stroke sensor **861** and the second stroke sensor **862** are normal, the processing proceeds to step S**470**. When the first stroke sensor **861** or the second stroke sensor **862** is abnormal, the second control calculation section **623** determines which of the first stroke sensor **861** or the second stroke sensor **862** is abnormal. Then, the processing proceeds to step S**485**.

In step S**470** subsequent to step S**465**, the second control calculation section **623** determines whether the first drive circuit **71** is normal. Specifically, the second control calculation section **623** determines whether the second control calculation section **623** has received the signal indicating that the first drive circuit **71** is abnormal, output by the first control calculation section **613** in step S**385**, which has been described above. When the first drive circuit **71** is normal, the processing proceeds to step S**480**. When the first drive circuit **71** is abnormal, the processing proceeds to step S**490**.

In step S**480** subsequent to step S**470**, the second control calculation section **623** controls the second actuator **20**, similarly to step S**180** executed by the control calculation section **633**, which has been described above. Specifically, the second control calculation section **623** performs the normal control, the ABS control, the VSC control, and the like. Then, the processing returns to step S**400**.

In step S**485**, when the second control calculation section **623** itself is abnormal, the second control calculation section **623** outputs the signal indicating that the second control calculation section **623** is abnormal to the first control calculation section **613**. At this time, the second control calculation section **623** outputs the signal indicating that the second control calculation section **623** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second control calculation section **623** is abnormal.

When the second drive circuit **72** is abnormal, the second control calculation section **623** outputs the signal indicating that the second drive circuit **72** is abnormal to the first control calculation section **613**. At this time, the second control calculation section **623** outputs the signal indicating that the second drive circuit **72** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second drive circuit **72** is abnormal.

When the first sensor power supply line **821** is abnormal, the second control calculation section **623** outputs a signal indicating that the first sensor power supply line **821** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first sensor power supply line **821** is abnormal through screen display, sound, light, and the like.

When the second sensor power supply line **822** is abnormal, the second control calculation section **623** outputs a signal indicating that the second sensor power supply line **822** is abnormal to the notification device (not illustrated). When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second sensor power supply line **822** is abnormal through screen display, sound, light, and the like.

When the first stroke sensor **861** is abnormal, the second control calculation section **623** outputs a signal indicating that the first stroke sensor **861** is abnormal to the notification device. When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the first stroke sensor **861** is abnormal through screen display, sound, light, and the like.

When the second stroke sensor **862** is abnormal, the second control calculation section **623** outputs a signal indicating that the second stroke sensor **862** is abnormal to the notification device. When the notification device has received this signal, the notification device notifies the driver of the vehicle **6** that the second stroke sensor **862** is abnormal through screen display, sound, light, and the like. After step S**485**, the processing proceeds to step S**490**.

Subsequently, in step S**490**, when the first power supply **401** and the second power supply **402** are abnormal, when the second control calculation section **623** itself is abnormal, or when the second drive circuit **72** is abnormal, the second control calculation section **623** cannot control the second drive circuit **72** normally. Therefore, the vehicle **6** is controlled to decelerate and stop in order to ensure the safety of the vehicle **6** by another calculation section or the like that is different from the second control calculation section **623**.

For example, in these cases, when the first control calculation section **613** and the first drive circuit **71** are normal, the first control calculation section **613** controls the first actuator **10** by controlling the first drive circuit **71**, similarly to step S**390** described above. As a result, the vehicle **6** safely decelerates and stops.

In these cases, when the first control calculation section **613** and the first drive circuit **71** are abnormal, the regenerative brake (not illustrated), the parking brake (not illustrated), and the like are controlled by another ECU that is different from the ECU **53**, similarly to step S**390** described above. As a result, the vehicle **6** safely decelerates and stops.

For example, when the first power supply **401**, the second power supply **402**, the second control calculation section **623**, and the second drive circuit **72** are normal while the first control calculation section **613** or the first drive circuit **71** is abnormal, the second control calculation section **623** causes the vehicle **6** to decelerate and stop.

Specifically, the second control calculation section **623** outputs the signal for driving the second actuator **20** to the second drive circuit **72**. The second drive circuit **72** drives the second actuator motor **30** based on the signal from the second control calculation section **623**. The second actuator motor **30** rotates based on the signal from the second drive circuit **72** to drive the second pump **224** and the third pump **274**.

At this time, the second pump **224** sucks the brake fluid stored in the first pressure adjustment reservoir **221**. The sucked brake fluid flows between the first differential pressure control valve **212** and each of the first pressure increase control valve **215** and the second pressure increase control valve **218** after the brake fluid flows through the first return-flow pipe line **223**. The brake fluid caused to flow by the second pump **224** flows to the left front wheel W/C **2** through the first pressure increase control valve **215**. The brake fluid caused to flow by the second pump **224** flows to the right front wheel W/C **3** through the second pressure increase control valve **218**.

At this time, the third pump **274** sucks the brake fluid stored in the second pressure adjustment reservoir **271**. The sucked brake fluid flows between the second differential pressure control valve **262** and each of the third pressure increase control valve **265** and the fourth pressure increase control valve **268** after the brake fluid flows through the second return-flow pipe line **273**. The brake fluid caused to flow by the third pump **274** flows to the right rear wheel W/C **5** through the third pressure increase control valve **265**. The brake fluid caused to flow by the third pump **274** flows to the left rear wheel W/C **4** through the fourth pressure increase control valve **268**.

Thus, each brake pad (not illustrated) comes into frictional contact with the corresponding brake disc. Therefore, each wheel that corresponds to each brake disc is decelerated, and thus the vehicle **6** decelerates. As a result, the vehicle **6** stops. After step S**490**, the processing returns to step S**400**.

In this manner, the processing of the second control calculation section **623** is executed.

Effects similar to those in the first embodiment are obtained also in the second embodiment. In the second embodiment, the vehicle brake system **1** includes the first microcomputer **61** and the second microcomputer **62**. With this configuration, even if one of the first microcomputer **61** or the second microcomputer **62** fails, the vehicle **6** can be caused to decelerate and stop safely by using the other normal one. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

In the second embodiment, the first control calculation section **613** determines whether the first sensor power supply line **821** is normal based on the first sensor output Vs**1** of the first stroke sensor **861**, V**1**, and V**2**. In addition, the second control calculation section **623** determines whether the second sensor power supply line **822** is normal based on the second sensor output Vs**2** of the second stroke sensor **862**, V**1**, and V**2**. Further, the first control calculation section **613** and the second control calculation section **623** determine whether the first stroke sensor **861** is normal based on the first sensor output Vs**1** of the first stroke sensor **861**, the first sensor threshold Vs_th**1**, and the second sensor threshold Vs_th**2**. The first control calculation section **613** and the second control calculation section **623** also determine whether the second stroke sensor **862** is normal based on the second sensor output Vs**2** of the second stroke sensor **862**, the first sensor threshold Vs_th**1**, and the second sensor threshold Vs_th**2**. With these determinations, a distinction can be made between the abnormality of the wiring lines and the abnormality of the sensors. As described above, V**1**, V**2**, the first sensor threshold Vs_th**1**, and the second sensor threshold Vs_th**2** have the relationship of $V1<Vs\_th1<Vs\_th2<V2$.

The first control calculation section **613** and the second control calculation section **623** determine whether the first stroke sensor **861** and the second stroke sensor **862** are normal based on the sensor output difference Vs**1**-Vs**2**. With this determination, the abnormality of the sensors is more likely to be detected.

The vehicle brake system **1** also includes the first stroke sensor **861** and the second stroke sensor **862**. With this configuration, even if one of the first stroke sensor **861** or the second stroke sensor **862** fails, the vehicle **6** can be caused to decelerate and stop safely by using the other normal one. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

Third Embodiment

In a third embodiment, the vehicle brake system **1** does not include the power supply switching circuit **403** while the vehicle brake system **1** includes two ECUs. Wiring lines of the first power supply **401** and the second power supply **402** are different. The others are similar to those of the second embodiment.

In the third embodiment, as illustrated in FIGS. **16** and **17**, the vehicle brake system **1** further includes a first ECU **51** and a second ECU **52**.

The first ECU **51** corresponds to a first hydraulic pressure control device, and controls the first actuator **10** by controlling the first actuator motor **13**. Specifically, the first ECU **51** includes the first microcomputer **61** and the first drive circuit **71** described above.

The second ECU **52** corresponds to a second hydraulic pressure control device, and controls the second actuator **20** by controlling the second actuator motor **30**. Specifically, the second ECU **52** includes the second microcomputer **62** and the second drive circuit **72** described above.

Here, the first power supply **401** supplies power to the first ECU **51**.

The first voltage sensor **451** outputs, to the first ECU **51**, a signal being in accordance with a voltage applied from the first power supply **401** to the first ECU **51**.

The second power supply **402** supplies power to the second ECU **52**.

The second voltage sensor **452** outputs, to the second ECU **52**, a signal being in accordance with a voltage applied from the second power supply **402** to the second ECU **52**.

The vehicle brake system **1** according to the third embodiment is configured as described above.

In the third embodiment, in step S**310** executed by the first control calculation section **613**, the first control calculation section **613** determines whether the first power supply **401** is normal. Specifically, the first control calculation section **613** determines whether the first voltage Vb**1** obtained in step S**300** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**. When the first voltage Vb**1** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**, the first power supply **401** is normal. Thus, the processing proceeds to step S**320**. When the first voltage Vb**1** is lower than the first voltage threshold Vb_th**1**, the first power supply **401** is abnormal. Thus, the processing proceeds to step S**390**. When the first voltage Vb**1** is higher than the second voltage threshold Vb_th**2**, the first power supply **401** is abnormal. Thus, the processing proceeds to step S**390**. The other processing is similar to that described above.

In step **S410** executed by the second control calculation section **623**, the second control calculation section **623** determines whether the second power supply **402** is normal. Specifically, the second control calculation section **623** determines whether the second voltage Vb**2** obtained in step **S400** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**. When the second voltage Vb**2** is equal to or higher than the first voltage threshold Vb_th**1** and equal to or lower than the second voltage threshold Vb_th**2**, the second power supply **402** is normal. Thus, the processing proceeds to step S**420**. When the second voltage Vb**2** is lower than the first voltage threshold Vb_th**1**, the second power supply **402** is abnormal. Thus, the processing proceeds to step S**490**. When the second voltage Vb**2** is higher than the second voltage threshold Vb_th**2**, the second power supply **402** is abnormal. Thus, the processing proceeds to step S**490**. The other processing is similar to that described above.

Effects similar to those in the second embodiment are obtained also in the third embodiment.

Fourth Embodiment

In a fourth embodiment, as illustrated in FIG. **18**, the vehicle brake device **80** further includes a third sensor power supply line **823**, a third sensor ground line **833**, a fourth sensor power supply line **824**, and a fourth sensor ground line **834**. The others are similar to those of the third embodiment.

The third sensor power supply line **823** is connected in a manner similar to the first sensor power supply line **821**, and is connected to the first ECU **51** and the first stroke sensor **861**.

The third sensor ground line **833** is connected in a manner similar to the first sensor ground line **831**, and is connected to the first ECU **51** and the first stroke sensor **861**.

The fourth sensor power supply line **824** is connected in a manner similar to the second sensor power supply line **822**, and is connected to the second ECU **52** and the second stroke sensor **862**.

The fourth sensor ground line **834** is connected in a manner similar to the second sensor ground line **832**, and is connected to the second ECU **52** and the second stroke sensor **862**.

Effects similar to those in the third embodiment are obtained also in the fourth embodiment. In the fourth embodiment, the vehicle brake device **80** further includes the third sensor power supply line **823**, the third sensor ground line **833**, the fourth sensor power supply line **824**, and the fourth sensor ground line **834**. With this configuration, even if any of the wiring lines described above is disconnected, power is supplied to the first stroke sensor **861** and the second stroke sensor **862** through the normal wiring lines. Therefore, the redundancy of the vehicle brake system **1** can be ensured, and thus the redundancy of the vehicle brake system **1** is improved.

Fifth Embodiment

In a fifth embodiment, the vehicle brake device **80** does not include the first sensor output line **841**, the second sensor output line **842**, the third sensor output line **843**, and the fourth sensor output line **844**, while the vehicle brake device **80** includes the fifth sensor output line **845** and the sixth sensor output line **846**. The others are similar to those of the third embodiment.

As illustrated in FIGS. **19** and **20**, the fifth sensor output line **845** is connected to the first ECU **51** and the first stroke sensor **861**.

The sixth sensor output line **846** is connected to the second ECU **52** and the second stroke sensor **862**.

In this case, in step S**300**, the first control calculation section **613** obtains the first sensor output Vs**1** that corresponds to the stroke amount X of the brake pedal **81** from the first stroke sensor **861** through the fifth sensor output line **845** and the first communication section **611**. In step S**300**, the first control calculation section **613** also obtains the second sensor output Vs**2** that corresponds to the stroke amount X of the brake pedal **81** by wirelessly communicating with the second ECU **52**. The other processing is similar to that described above.

In step S**400**, the second control calculation section **623** obtains the second sensor output Vs**2** that corresponds to the stroke amount X of the brake pedal **81** from the second stroke sensor **862** through the sixth sensor output line **846** and the second communication section **621**. In step S**400**, the second control calculation section **623** also obtains the first sensor output Vs**1** that corresponds to the stroke amount X of the brake pedal **81** by wirelessly communicating with the first ECU **51**. The other processing is similar to that described above.

Effects similar to those in the third embodiment are obtained also in the fifth embodiment. In the fifth embodiment, the number of wiring lines can be reduced as compared to that in the third embodiment. Thus, weight of the vehicle brake device **80** and cost such as material cost can be reduced.

Sixth Embodiment

In a sixth embodiment, the vehicle brake system **1** does not include the power supply switching circuit **403**, the first voltage sensor **451**, and the second voltage sensor **452** while the vehicle brake system **1** includes a power distribution unit **404** as illustrated in FIG. **21**. The others are similar to those of the second embodiment.

The power distribution unit **404** receives power supply from the first power supply **401** and the second power supply **402**. The power distribution unit **404** includes a power distribution circuit (not illustrated), and receives power supplied from the first power supply **401** and the second power supply **402**. The power distribution unit **404** distributes the received power to the first microcomputer **61** and the second microcomputer **62**.

The power distribution unit **404** further includes a microcomputer (not illustrated), a ROM (not illustrated), a voltage sensor (not illustrated), and the like. When the ignition of the vehicle **6** is turned on, the power distribution unit **404** distributes the power supplied from the first power supply **401** and the second power supply **402** to the first microcomputer **61** and the second microcomputer **62**. At this time, the power distribution unit **404** determines whether the first power supply **401** and the second power supply **402** are normal by executing a program stored in the ROM of the power distribution unit **404**. Hereinafter, the determination of the power distribution unit **404** will be described with reference to a flowchart of FIG. **22**.

In step S**500**, the power distribution unit **404** determines whether the first power supply **401** and the second power supply **402** are normal. Specifically, the voltage sensor (not illustrated) of the power distribution unit **404** measures a voltage applied from the first power supply **401** to the power distribution unit **404**. The voltage sensor (not illustrated) of the power distribution unit **404** also measures a voltage applied from the second power supply **402** to the power distribution unit **404**. Then, the power distribution unit **404** determines whether each of these measured voltages is equal to or higher than a first distribution-used voltage threshold and equal to or lower than a second distribution-used voltage threshold. When each of the measured voltages is equal to or higher than the first distribution-used voltage threshold and equal to or lower than the second distribution-used voltage threshold, the first power supply **401** and the second power supply **402** are normal. Thus, the processing proceeds to step S**510**. When one of the measured voltages is lower than the first distribution-used voltage threshold, the first power supply **401** or the second power supply **402** is abnormal. Thus, the processing proceeds to step S**520**. When one of the measured voltages is higher than the second distribution-used voltage threshold, the first power supply **401** or the second power supply **402** is abnormal. Thus, the processing proceeds to step S**520**.

In step S**510** subsequent to step S**500**, the power distribution unit **404** transmits a normality signal indicating that the first power supply **401** and the second power supply **402** are normal to the first microcomputer **61** and the second microcomputer **62**. Then, the processing returns to step S**500**.

In step S**520** subsequent to step S**500**, the power distribution unit **404** transmits an abnormality signal indicating that the first power supply **401** or the second power supply **402** is abnormal (for example, the first power supply **401** or the second power supply **402** is disconnected) to the first microcomputer **61** and the second microcomputer **62**. Then, the processing returns to step S**500**.

Then, in step S**310**, the first control calculation section **613** determines whether the first power supply **401** and the second power supply **402** are normal based on the normality signal and the abnormality signal from the power distribution unit **404**. The other processing is similar to that described above.

In step S**410**, the second control calculation section **623** determines whether the first power supply **401** and the second power supply **402** are normal based on the normality signal and the abnormality signal from the power distribution unit **404**. The other processing is similar to that described above.

In this manner, the processing of the power distribution unit **404**, the first control calculation section **613**, and the second control calculation section **623** is executed.

Effects similar to those in the second embodiment are obtained also in the sixth embodiment.

Seventh Embodiment

In a seventh embodiment, the pressure of the brake fluid from the reservoir **11** is increased without using the first pump **12**. The others are similar to those of the first embodiment.

As illustrated in FIG. **23**, the first actuator **10** includes a gear mechanism **15**, an actuator cylinder **16**, and an actuator piston **17** in addition to the reservoir **11**, the first actuator motor **13**, and the first pressure sensor **14** described above.

The gear mechanism **15** has a ball screw and a rack and pinion. The gear mechanism **15** is connected to the first actuator motor **13**. Thus, the gear mechanism **15** is caused to perform translational movement by rotation of the first actuator motor **13**.

The actuator cylinder **16** is formed in a bottomed cylindrical shape, and is connected to the reservoir **11**. Thus, the brake fluid in the reservoir **11** flows into the actuator cylinder **16** through a hole (not illustrated) of the reservoir **11** and a hole (not illustrated) of the actuator cylinder **16**.

The actuator piston **17** is connected to the gear mechanism **15**, and has a plurality of actuator springs **171**. With this configuration, when the gear mechanism **15** is caused to translate by the rotation of the first actuator motor **13**, the actuator piston **17** is caused to perform translational movement along with the gear mechanism **15**. As a result, the actuator piston **17** slides in the actuator cylinder **16** along an axial direction of the actuator cylinder **16**. When the actuator piston **17** slides, a pressure of the brake fluid in the actuator cylinder **16** is increased. Simultaneously, another hole (not illustrated) of the actuator cylinder **16**, which is connected to the first main pipe line **211** and the second main pipe line **261**, is opened. At this time, the brake fluid with the increased hydraulic pressure flows from the other hole (not illustrated) of the actuator cylinder **16** toward the second actuator **20**.

When the power supply is stopped from the first drive circuit **71** to the first actuator motor **13**, the first actuator motor **13** stops its rotation. When the rotation of the first actuator motor **13** is stopped, the actuator piston **17** is caused to perform the translational movement along with the gear mechanism **15** by a restoration force of the plurality of actuator springs **171**. As a result, the actuator piston **17** returns to the initial position.

Effects similar to those in the first embodiment are obtained also in the seventh embodiment.

Eighth Embodiment

In an eighth embodiment, as illustrated in FIG. **24**, the first power supply **401** supplies power to the first ECU **51** and the second ECU **52**. The second power supply **402** supplies power to the first ECU **51** and the second ECU **52**. The others are similar to those of the third embodiment.

Effects similar to those in the third embodiment are obtained also in the eighth embodiment.

OTHER EMBODIMENTS

The present disclosure is not limited to the above embodiments, and may be appropriately modified from the above embodiments. In the above embodiments, it goes without saying that the constituent elements forming the embodiments are not necessarily indispensable unless otherwise clearly stated or unless otherwise thought to be clearly indispensable in principle.

Each control unit and the like, and each method thereof described in the present disclosure may be implemented by a dedicated computer provided by including a processor and a memory programmed to execute one or more functions embodied by a computer program. Alternatively, each control unit and the like, and each method thereof described in the present disclosure may be implemented by a dedicated computer provided by including a processor with one or more dedicated hardware logic circuits. Alternatively, each control unit and the like, and each method thereof described in the present disclosure may be implemented by one or more dedicated computers configured by a combination of a processor and a memory programmed to execute one or more functions, and a processor with one or more hardware logic circuits. The computer program may be stored in a computer-readable non-transitory tangible storage medium as an instruction to be executed by the computer.

(1) In the above embodiments, the vehicle brake system **1** includes the first power supply **401** and the second power supply **402**. With regard to this configuration, the number of power supplies is not limited to one or two, and may be three or more.

(2) In the above embodiments, the vehicle brake system **1** includes the microcomputer **63**. Alternatively, the vehicle brake system **1** includes the first microcomputer **61** and the second microcomputer **62**. The number of microcomputers is not limited to one or two, and may be three or more.

(3) In the above embodiments, the vehicle brake system **1** includes the first actuator **10**, which corresponds to the first hydraulic pressure generation unit, and the second actuator **20**, which corresponds to the second hydraulic pressure generation unit. The number of hydraulic pressure generation units is not limited to two, and may be three or more.

(4) In the above embodiments, the vehicle brake system **1** includes the first drive circuit **71** and the second drive circuit **72**. The number of drive circuits is not limited to two, and may be one, or three or more.

(5) In the above embodiments, the vehicle brake device **80** includes the sensor power supply line **82**, the sensor ground line **83**, the first sensor output line **841**, and the second sensor output line **842**. The number of wiring lines for each type is not limited to one or two, and may be three or more.

(6) In the above embodiments, the vehicle brake device **80** includes the stroke sensor **86**. Alternatively, the vehicle brake device **80** includes the first stroke sensor **861** and the second stroke sensor **862**. With regard to these configurations, the number of stroke sensors is not limited to one or two, and may be three or more.

(7) In the above embodiments, the reaction force generation portion **90** of the vehicle brake device **80** includes the elastic member **91**. With regard to this configuration, the number of elastic members **91** is not limited to one, and may be two or more.

(8) The first to eighth embodiments may be appropriately combined.

As illustrated in FIG. **25**, in addition to the driving of the first actuator **10**, the first drive circuit **71** may drive the second actuator **20** by supplying power to the second actuator motor **30** based on a signal from the control calculation section **633**. In addition to the driving of the second actuator **20**, the second drive circuit **72** may drive the first actuator **10** by supplying power to the first actuator motor **13** based on a signal from the control calculation section **633**.

(9) In the above embodiments, the elastic member **91** is the helical compression spring. Unlike this configuration, the elastic member **91** may be a tension spring. In addition, the elastic member **91** may not be limited to an equal pitch spring, and may be a conical spring, an unequal pitch spring, or the like.

What is claimed is:

1. A vehicle brake system comprising:

a vehicle brake device including
- a brake pedal that has a pedal portion, and a lever portion rotating about a rotation axis in response to the pedal portion being operated,
- a first stroke sensor configured to output a first signal based on a stroke amount of the brake pedal,
- a second stroke sensor configured to output a second signal based on the stroke amount of the brake pedal,
- a housing that rotatably supports the lever portion, and
- a reaction force generation portion configured to generate a reaction force applied on the lever portion based on the stroke amount;

a first hydraulic pressure generation unit configured to generate a hydraulic pressure for braking a vehicle;

a second hydraulic pressure generation unit configured to generate hydraulic pressure for braking the vehicle;

a first hydraulic pressure control device that includes a first drive circuit configured to drive the first hydraulic pressure generation unit;

a second hydraulic pressure control device that includes a second drive circuit configured to drive the second hydraulic pressure generation unit;

a first power supply configured to supply power to the first hydraulic pressure control device;

a second power supply configured to supply power to the second hydraulic pressure control device;

a power supply switching circuit configured to switch a power supply source that supplies the power to the first hydraulic pressure control device and the second hydraulic pressure control device, between the first power supply and the second power supply based on a first voltage that is a voltage applied from the first power supply to the first hydraulic pressure control device and a second voltage that is a voltage applied from the second power supply to the second hydraulic pressure control device;

a first sensor power supply line that is connected to the first hydraulic pressure control device and the first stroke sensor, and configured to supply the power from the first power supply, from the first hydraulic pressure control device to the first stroke sensor;

a second sensor power supply line that is connected to the second hydraulic pressure control device and the second stroke sensor, and configured to supply the power from the second power supply, from the second hydraulic pressure control device to the second stroke sensor;

a third sensor power supply line that is connected to the first hydraulic pressure control device and the first stroke sensor, and configured to supply the power from the first power supply, from the first hydraulic pressure control device to the first stroke sensor;

a fourth sensor power supply line that is connected to the second hydraulic pressure control device and the second stroke sensor, and configured to supply the power from the second power supply, from the second hydraulic pressure control device to the second stroke sensor;

a first sensor output line that is connected to the first hydraulic pressure control device and the first stroke sensor, and configured to transmit the first signal from the first stroke sensor to the first hydraulic pressure control device;

a second sensor output line that is connected to the first hydraulic pressure control device and the second stroke sensor, and configured to transmit the second signal from the second stroke sensor to the first hydraulic pressure control device;

a third sensor output line that is connected to the second hydraulic pressure control device and the first stroke sensor, and configured to transmit the first signal from the first stroke sensor to the second hydraulic pressure control device;

a fourth sensor output line that is connected to the second hydraulic pressure control device and the second stroke sensor, and configured to transmit the second signal from the second stroke sensor to the second hydraulic pressure control device,

wherein

the first hydraulic pressure control device is configured to determine whether the first stroke sensor and the second stroke sensor are normal based on the first signal from the first stroke sensor transmitted through the first sensor output line and the second signal from the second stroke sensor transmitted through the second sensor output line, and when the first stroke sensor and the second stroke sensor are normal, control the first drive circuit based on at least one of the first signal from the first stroke sensor or the second signal from the second stroke sensor to control the hydraulic pressure generated by the first hydraulic pressure generation unit, and

the second hydraulic pressure control device is configured to determine whether the first stroke sensor and the second stroke sensor are normal based on the first signal from the first stroke sensor transmitted through the third sensor output line and the second signal from the second stroke sensor transmitted through the fourth sensor output line, and when the first stroke sensor and the second stroke sensor are normal, control the second drive circuit based on at least one of the first signal from the first stroke sensor or the second signal from the second stroke sensor to control the hydraulic pressure generated by the second hydraulic pressure generation unit.

2. The vehicle brake system according to claim **1**, wherein

the first and second hydraulic pressure control devices includes

a first hydraulic pressure control unit configured to control the hydraulic pressure generated by the first hydraulic pressure generation unit, by controlling the first drive circuit based on the first signal from the first stroke sensor, and

a second hydraulic pressure control unit configured to control the hydraulic pressure generated by the second hydraulic pressure generation unit, by controlling the second drive circuit based on the second signal from the second stroke sensor.

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